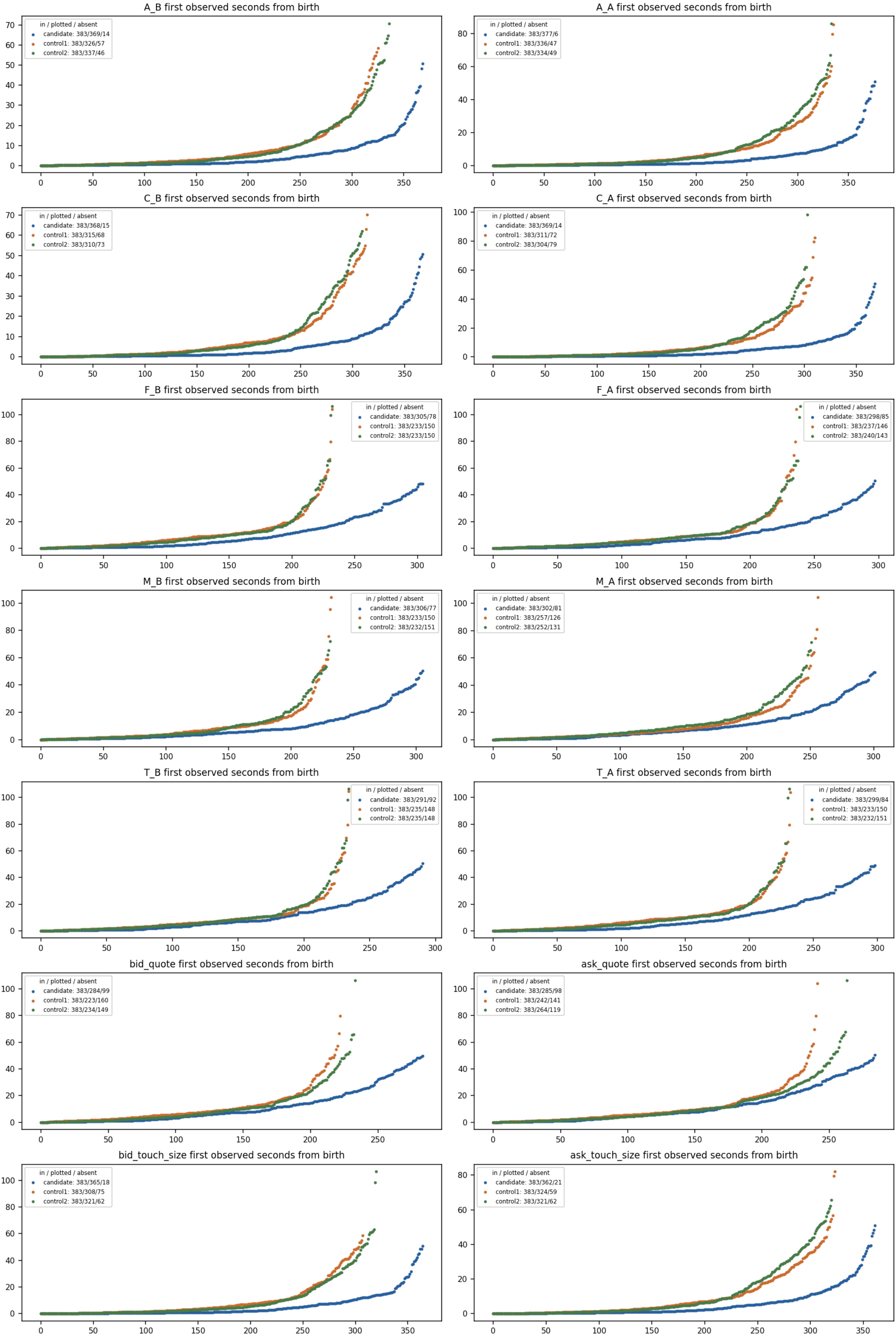
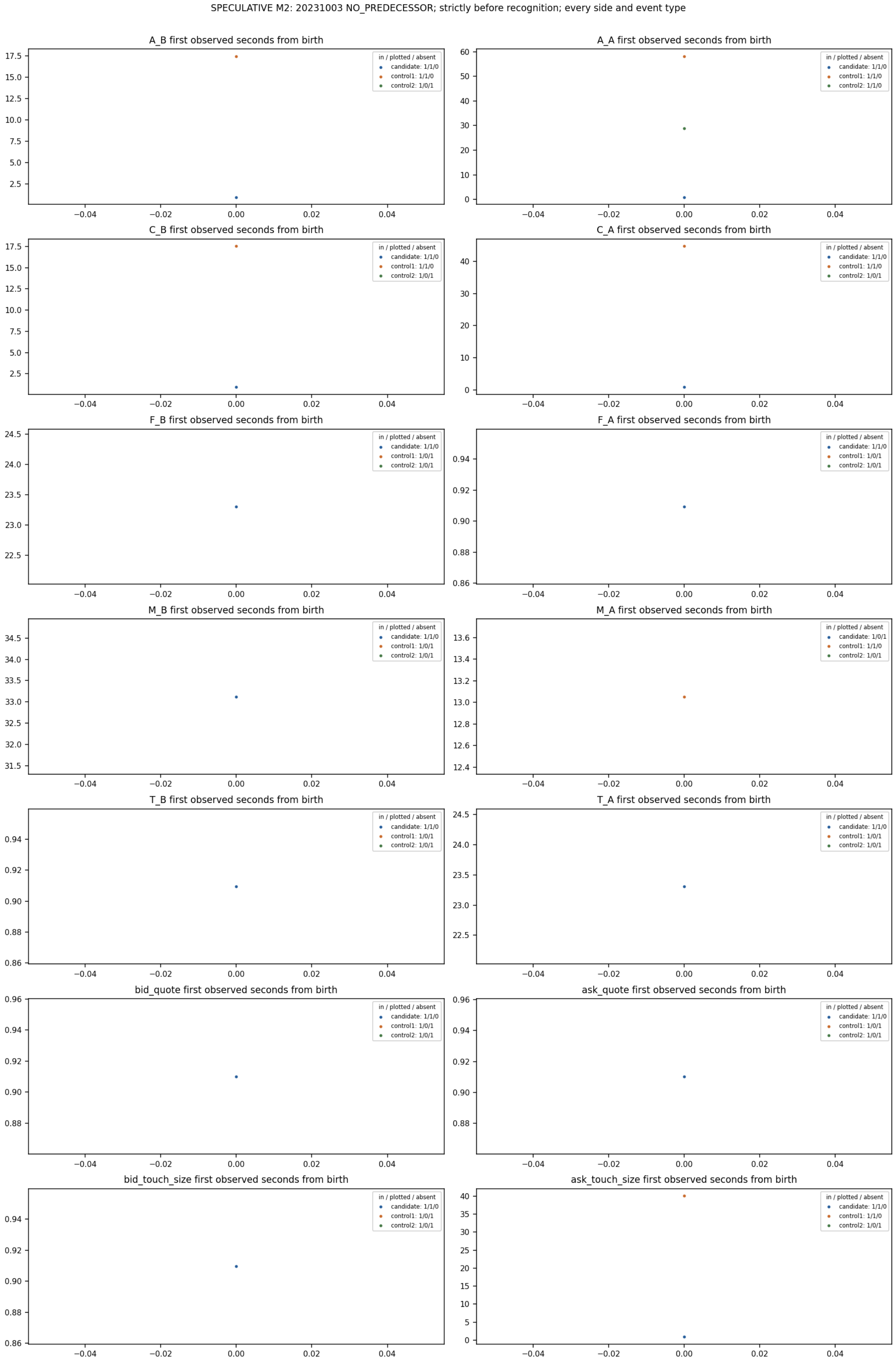
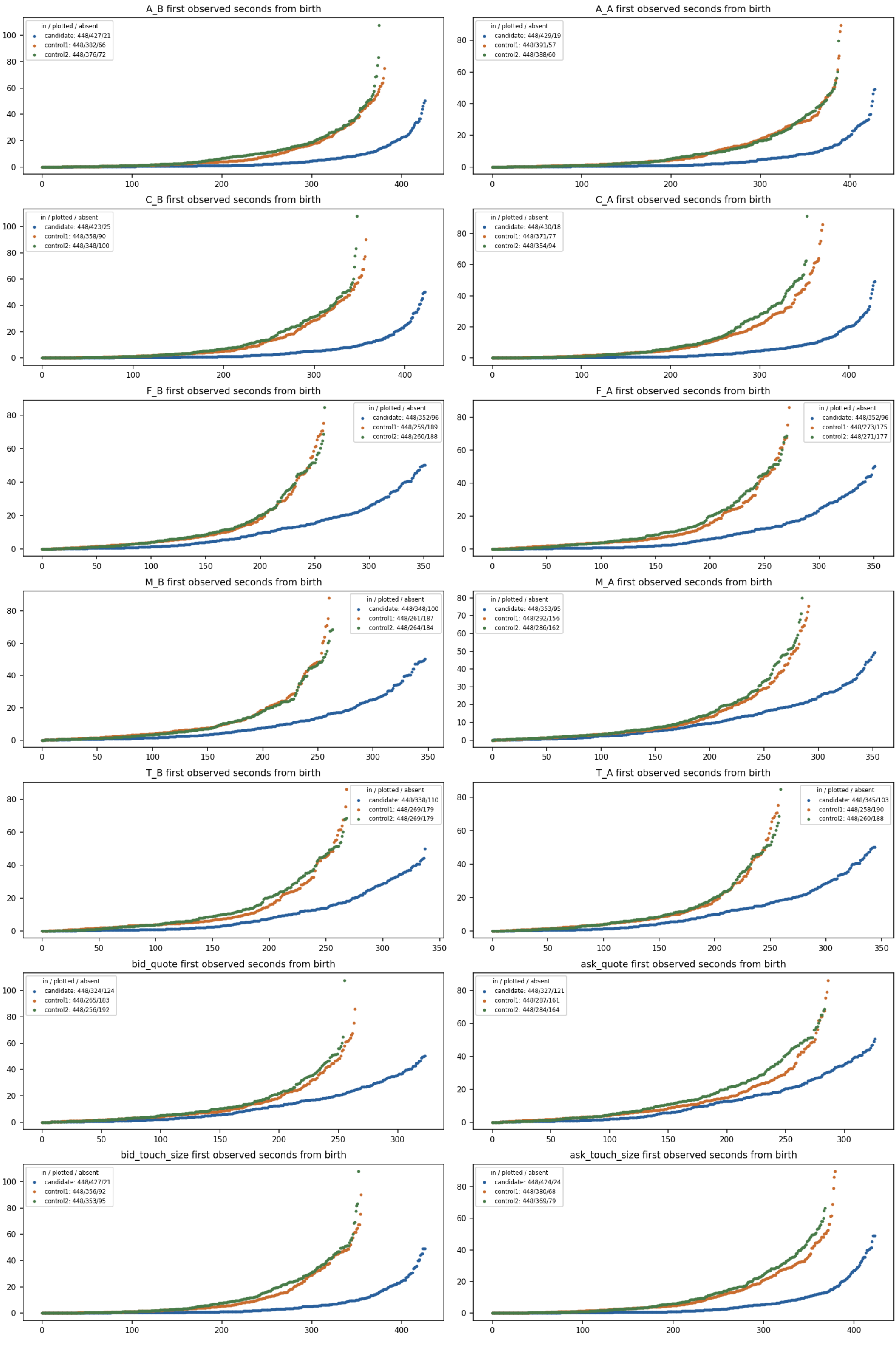
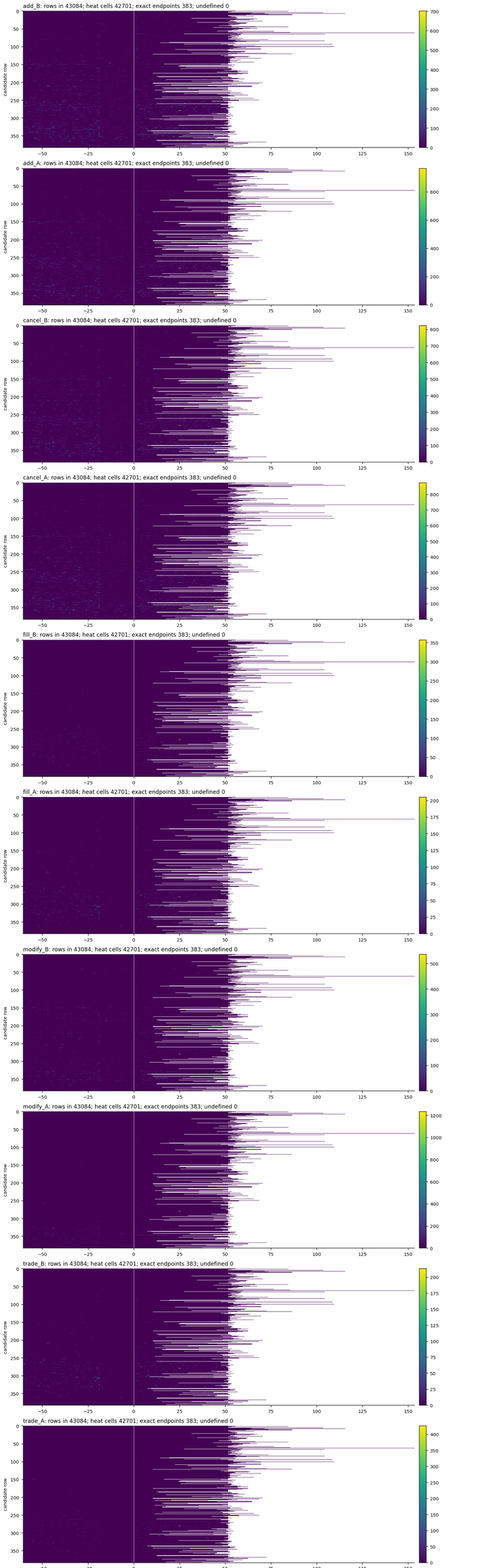

SPECULATIVE M2: 20231003 FLIP; strictly before recognition; every side and event type
A_B first observed seconds from birth
in / plotted / absent
candidate: 383/369/14
control1: 383/326/57
control2: 383/337/46
A_A first observed seconds from birth
in / plotted / absent
candidate: 383/377/6
control1: 383/336/47
control2: 383/334/49
C_B first observed seconds from birth
in / plotted / absent
candidate: 383/368/15
control1: 383/315/68
control2: 383/310/73
C_A first observed seconds from birth
in / plotted / absent
candidate: 383/369/14
control1: 383/311/72
control2: 383/304/79
F_B first observed seconds from birth
in / plotted / absent
candidate: 383/305/78
control1: 383/233/150
control2: 383/233/150
F_A first observed seconds from birth
in / plotted / absent
candidate: 383/298/85
control1: 383/237/146
control2: 383/240/143
M_B first observed seconds from birth
in / plotted / absent
candidate: 383/306/77
control1: 383/233/150
control2: 383/232/151
M_A first observed seconds from birth
in / plotted / absent
candidate: 383/302/81
control1: 383/257/126
control2: 383/252/131
T_B first observed seconds from birth
in / plotted / absent
candidate: 383/291/92
control1: 383/235/148
control2: 383/235/148
T_A first observed seconds from birth
in / plotted / absent
candidate: 383/299/84
control1: 383/233/150
control2: 383/232/151
bid_quote first observed seconds from birth
in / plotted / absent
candidate: 383/284/99
control1: 383/223/160
control2: 383/234/149
ask_quote first observed seconds from birth
in / plotted / absent
candidate: 383/285/98
control1: 383/242/141
control2: 383/264/119
bid_touch_size first observed seconds from birth
in / plotted / absent
candidate: 383/365/18
control1: 383/308/75
control2: 383/321/62
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 383/362/21
control1: 383/324/59
control2: 383/321/62


SPECULATIVE M2: 20231003 NO_PREDECESSOR; strictly before recognition; every side and event type
A_B first observed seconds from birth
A_A first observed seconds from birth
C_B first observed seconds from birth
C_A first observed seconds from birth
F_B first observed seconds from birth
F_A first observed seconds from birth
M_B first observed seconds from birth
M_A first observed seconds from birth
T_B first observed seconds from birth
T_A first observed seconds from birth
bid_quote first observed seconds from birth
ask_quote first observed seconds from birth
bid_touch_size first observed seconds from birth
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 1/1/0
control1: 1/1/0
control2: 1/0/1
control1: 1/0/1
control2: 1/0/1
candidate: 1/0/1

SPECULATIVE M2: 20231003 SAME; strictly before recognition; every side and event type

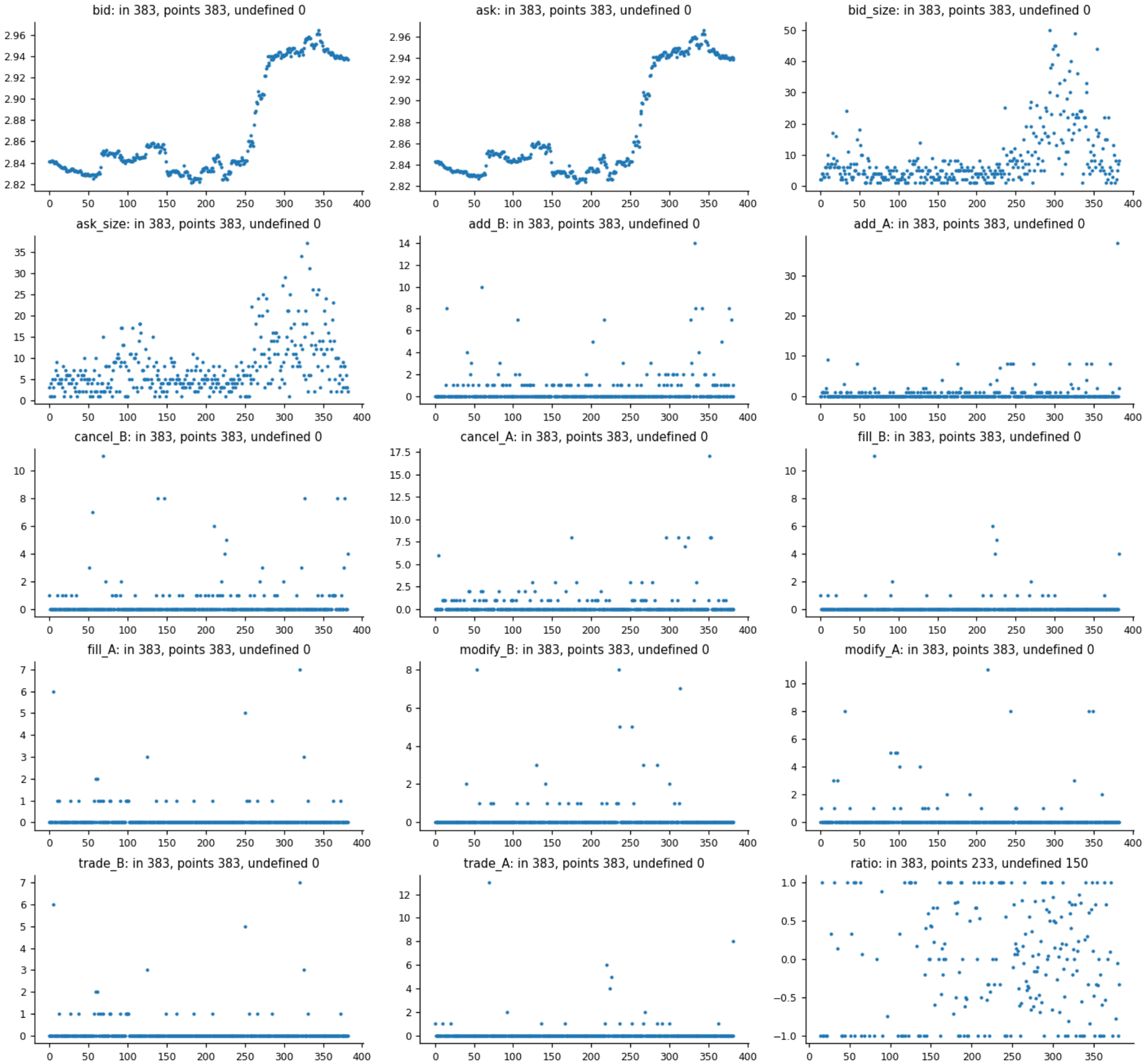


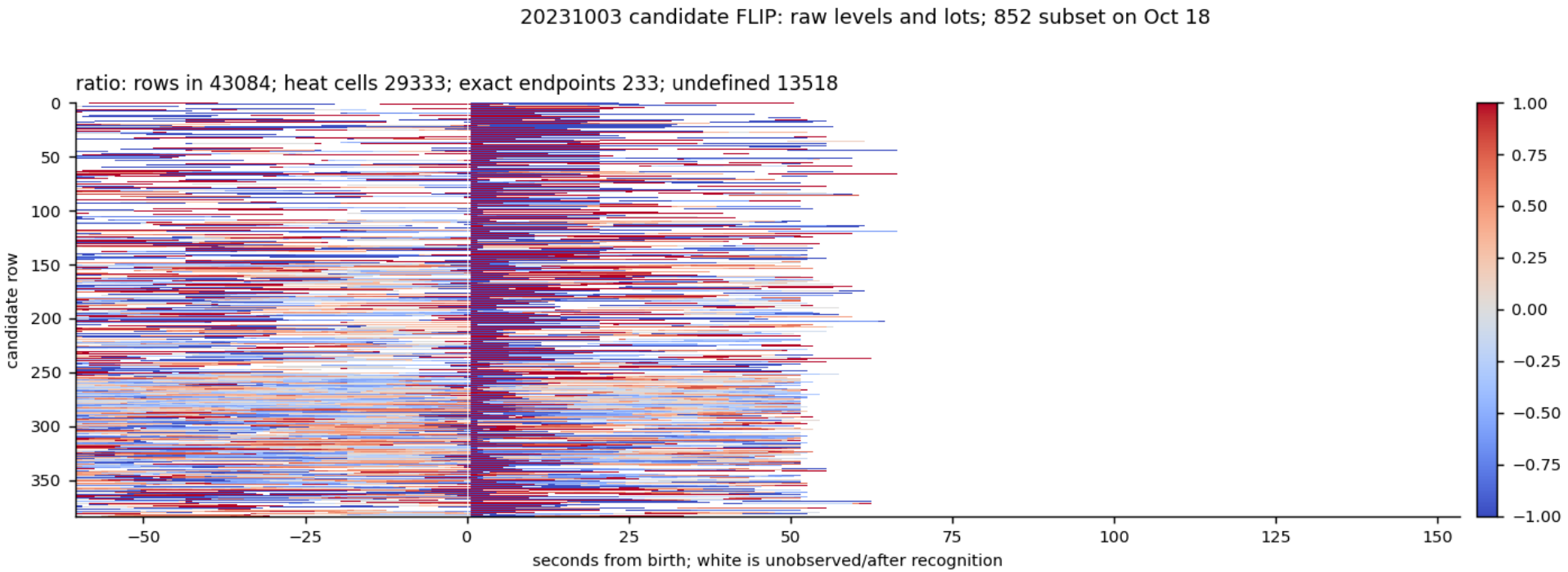
20231003 candidate FLIP: raw levels and lots; 852 subset on Oct 18



20231003-candidate-FLIP-endpoints.png

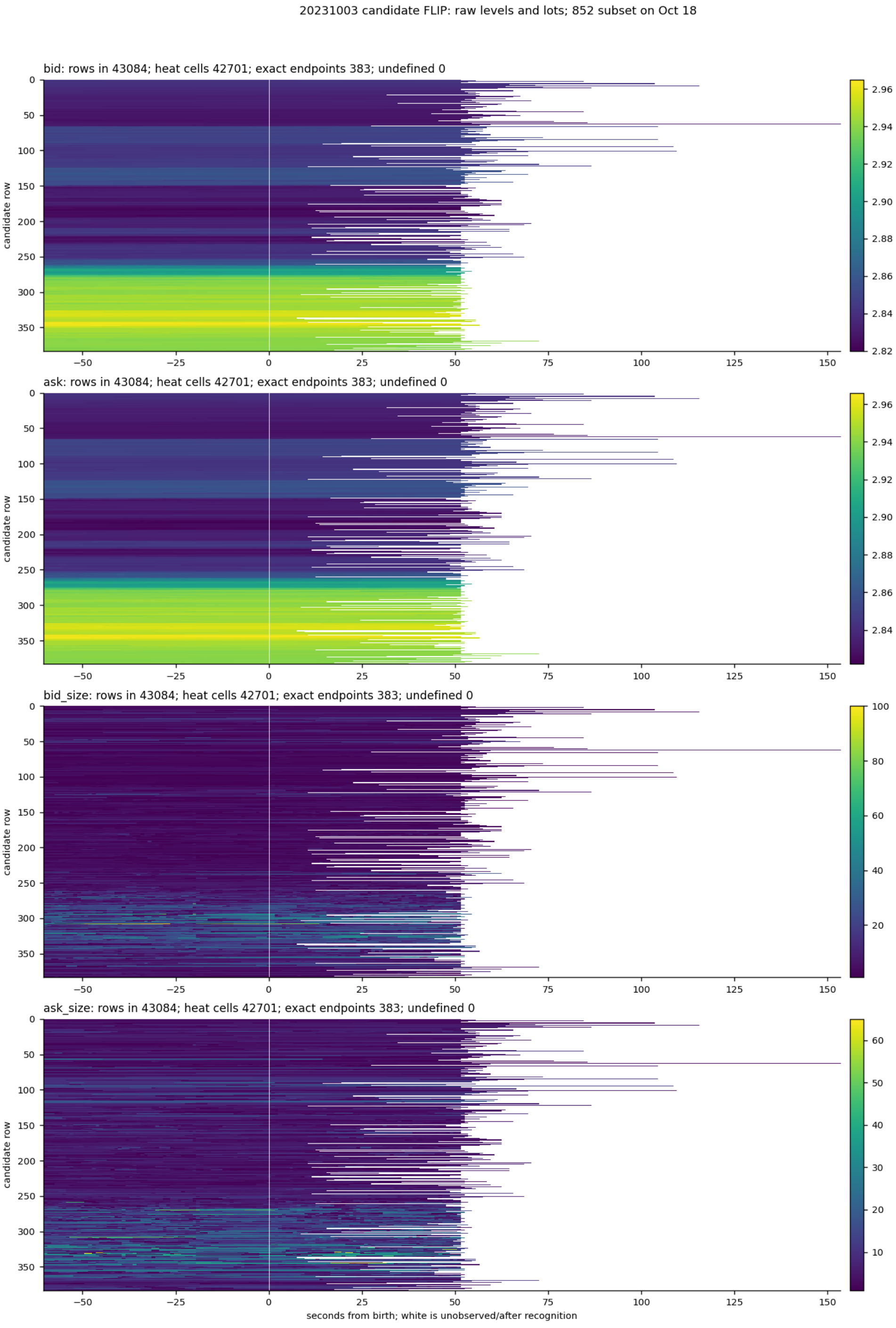
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231003 candidate FLIP: raw levels and lots; 852 subset on Oct 18
ratio: rows in 43084; heat cells 29333; exact endpoints 233; undefined 13518
candidate row
0
50
100
150
200
250
300
350
−50
−25
0
25
50
75
100
125
150
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

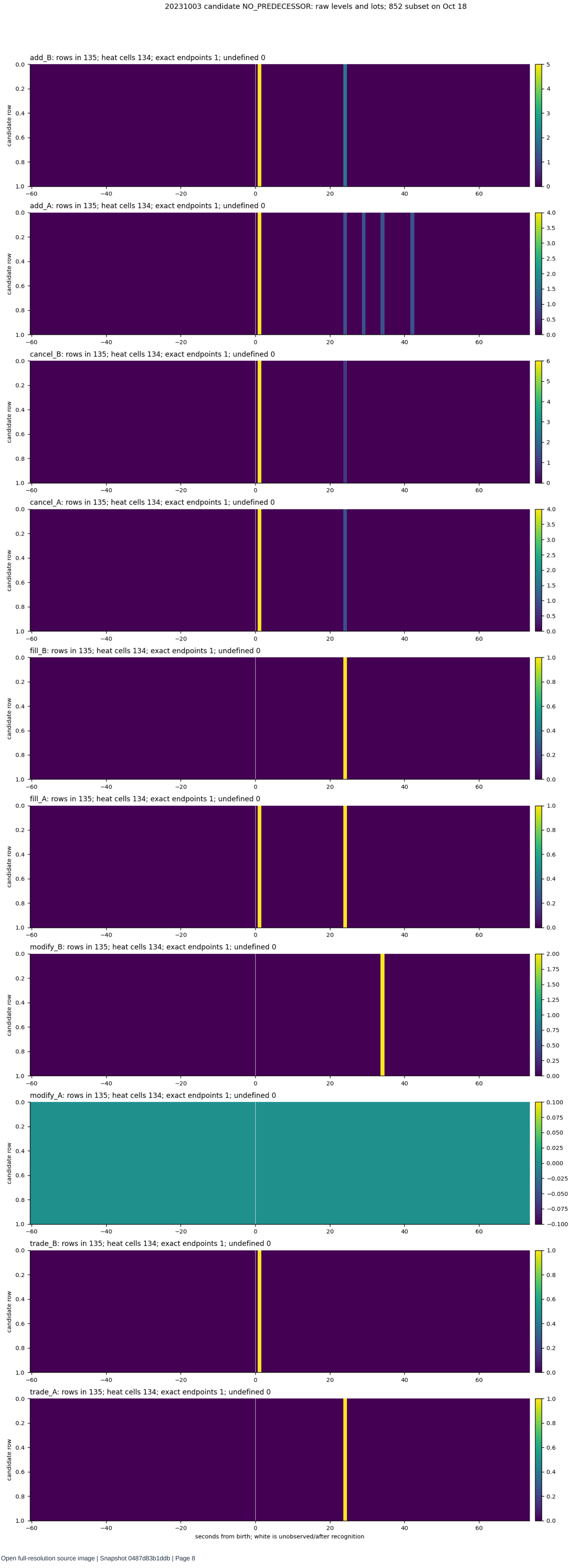
20231003-candidate-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



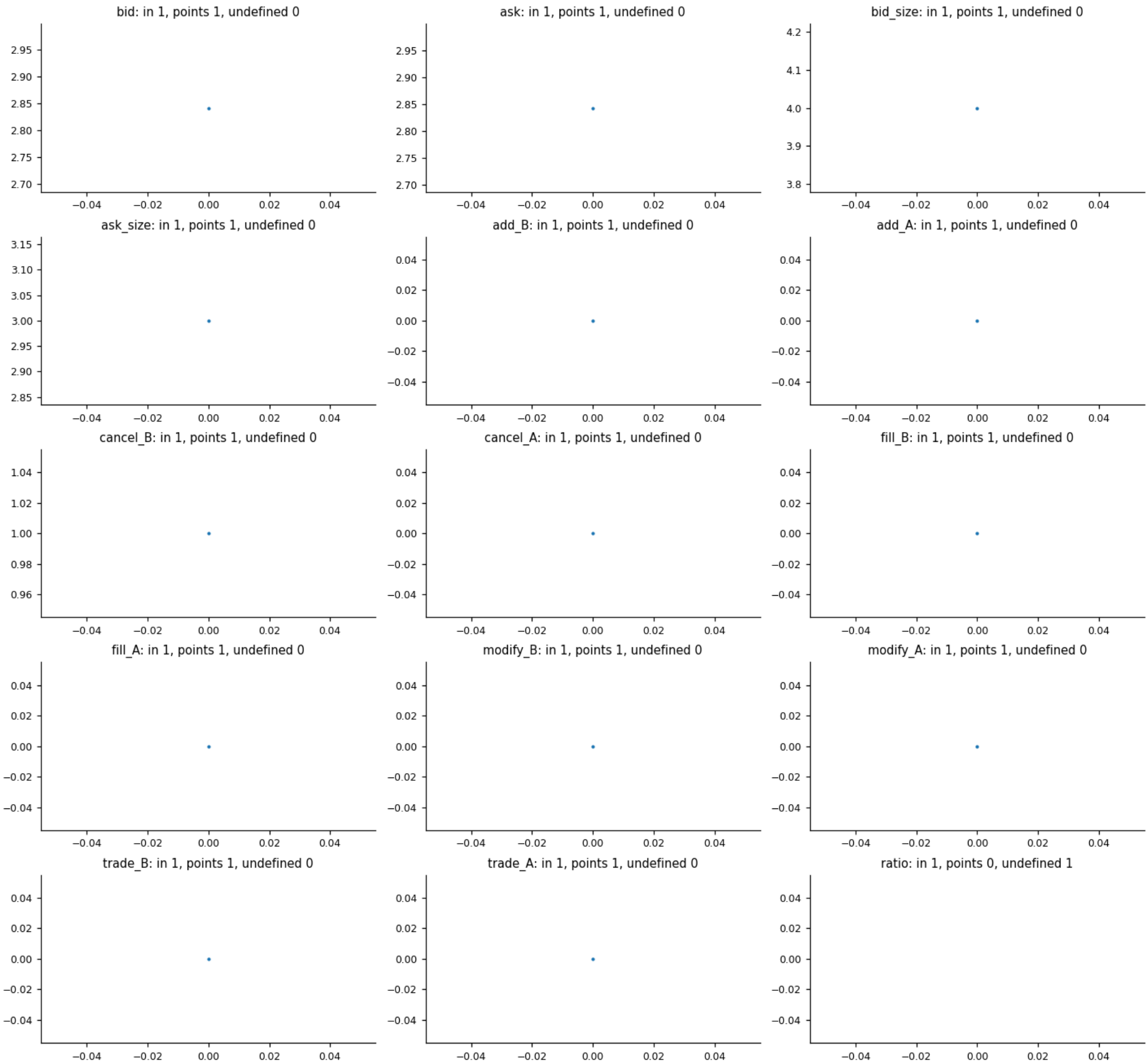
20231003-candidate-NO_PREDECESSOR-actions.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

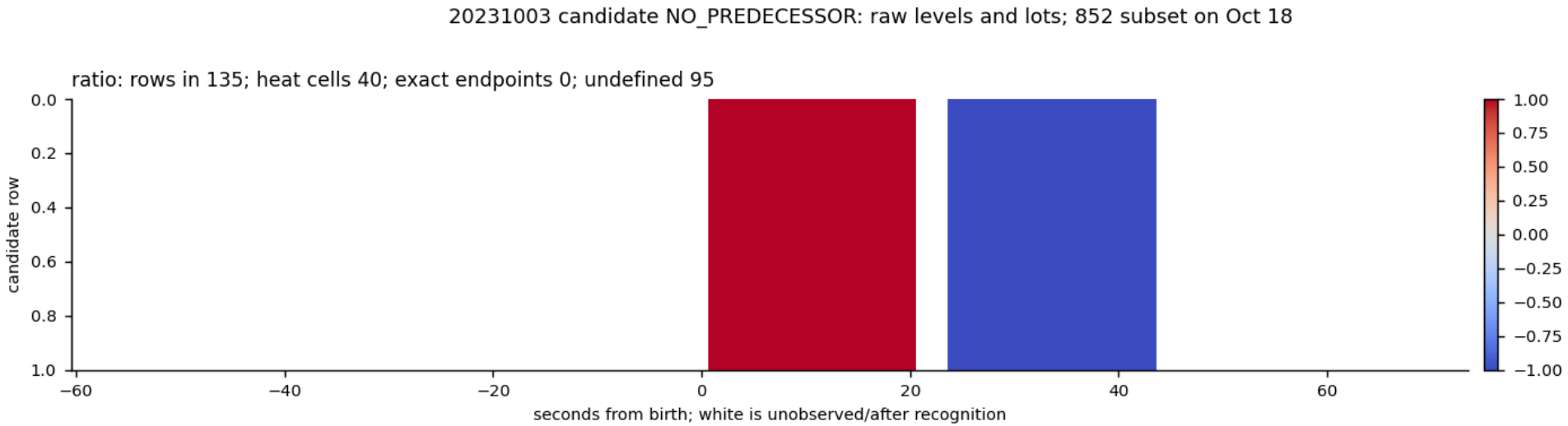


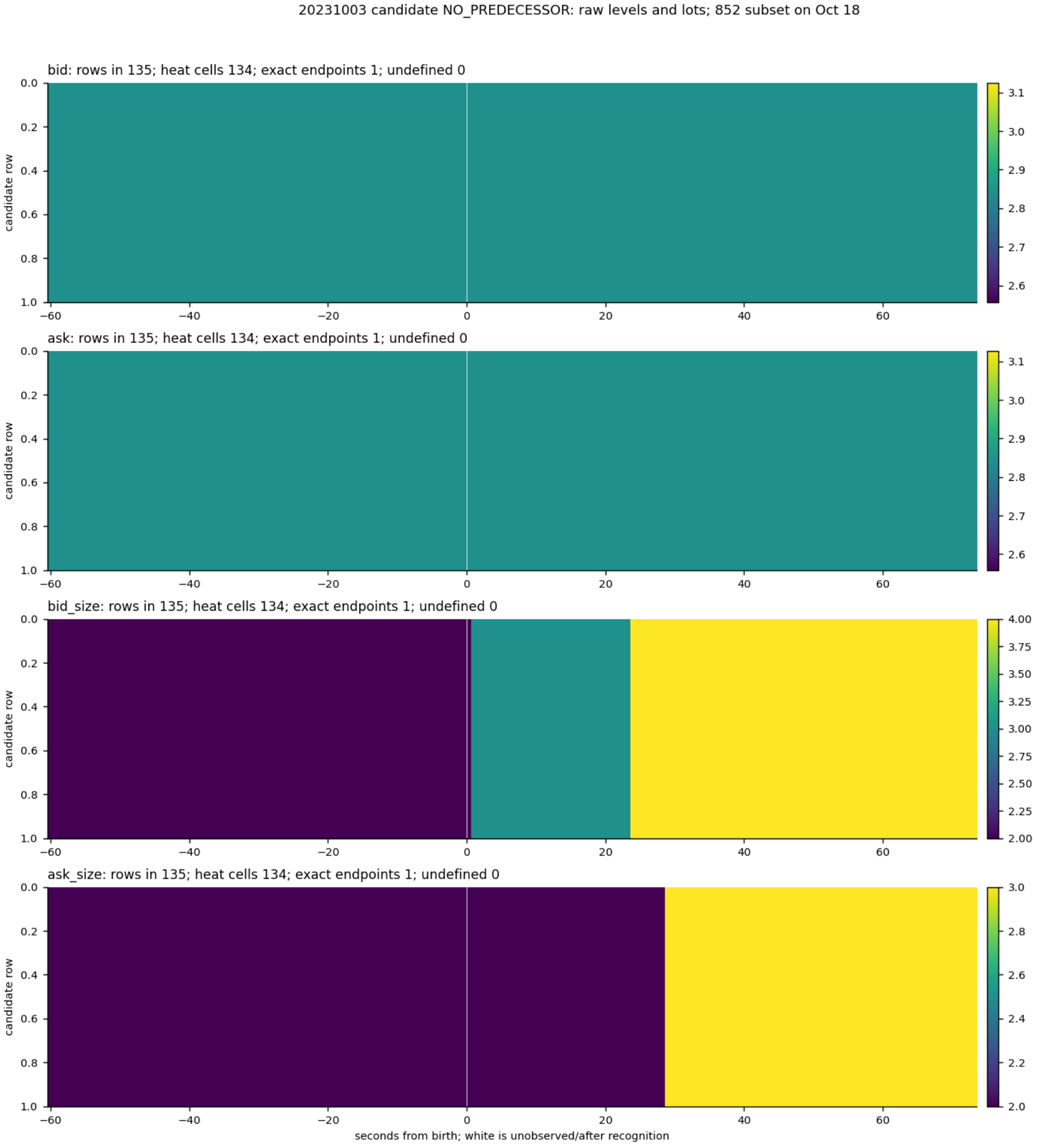
20231003-candidate-NO_PREDECESSOR-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

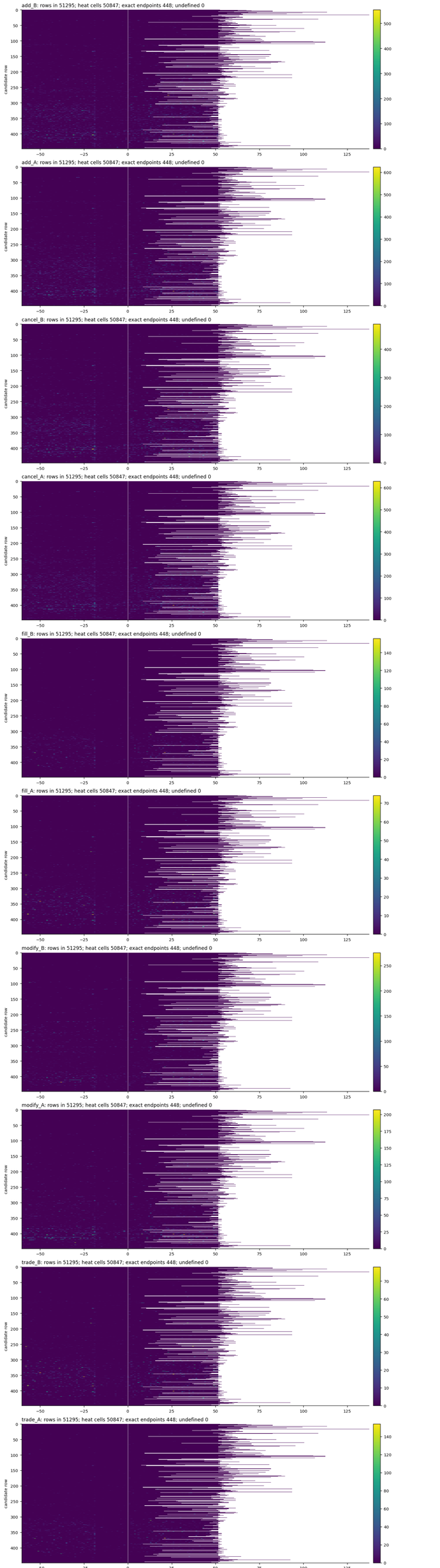


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



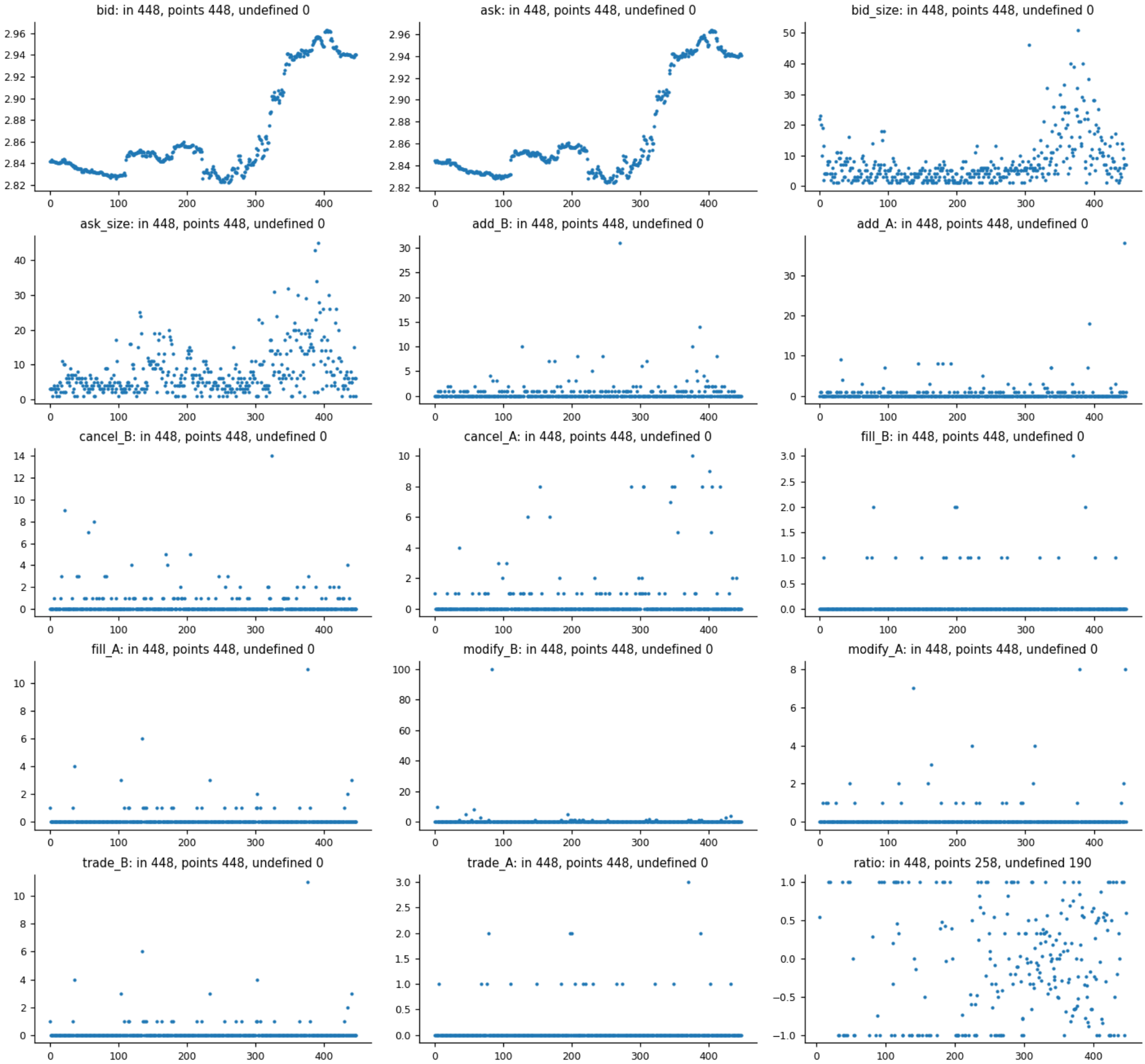

20231003 candidate NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 135; heat cells 134; exact endpoints 1; undefined 0
ask: rows in 135; heat cells 134; exact endpoints 1; undefined 0
bid_size: rows in 135; heat cells 134; exact endpoints 1; undefined 0
ask_size: rows in 135; heat cells 134; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

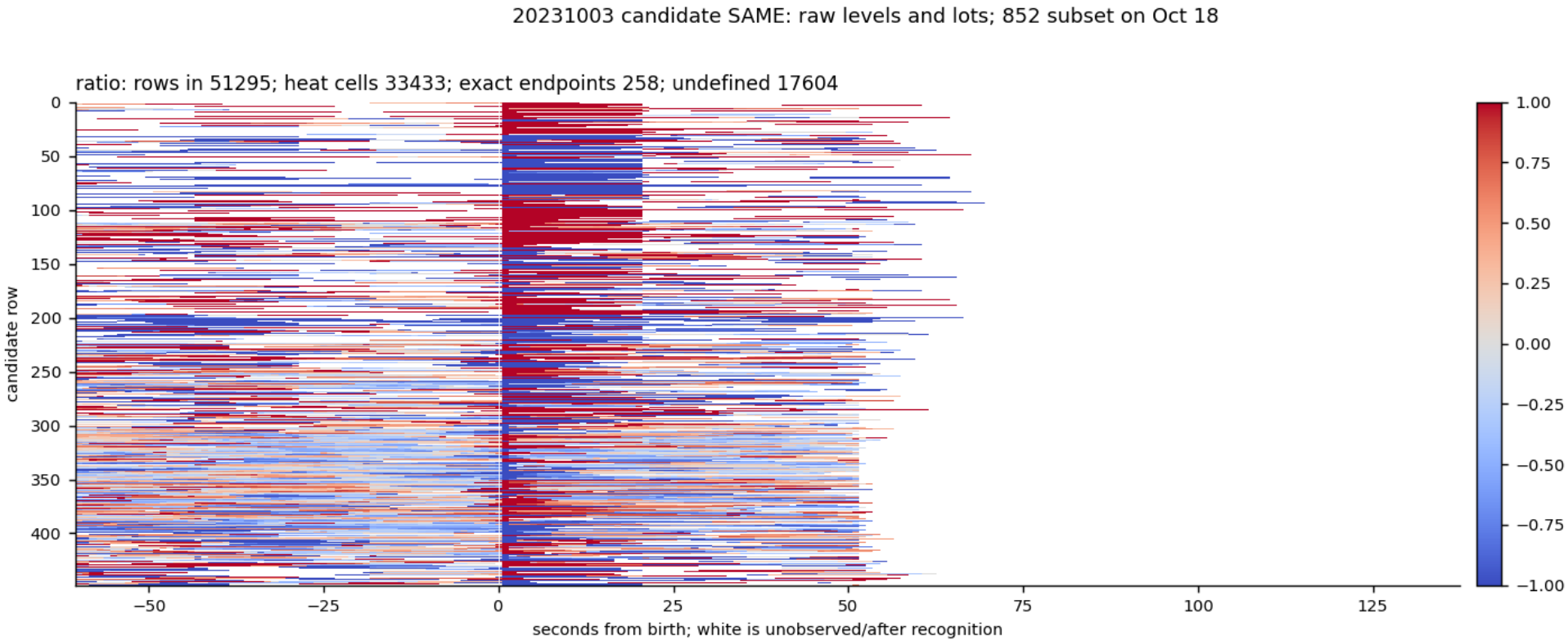
20231003 candidate SAME: raw levels and lots; 852 subset on Oct 18

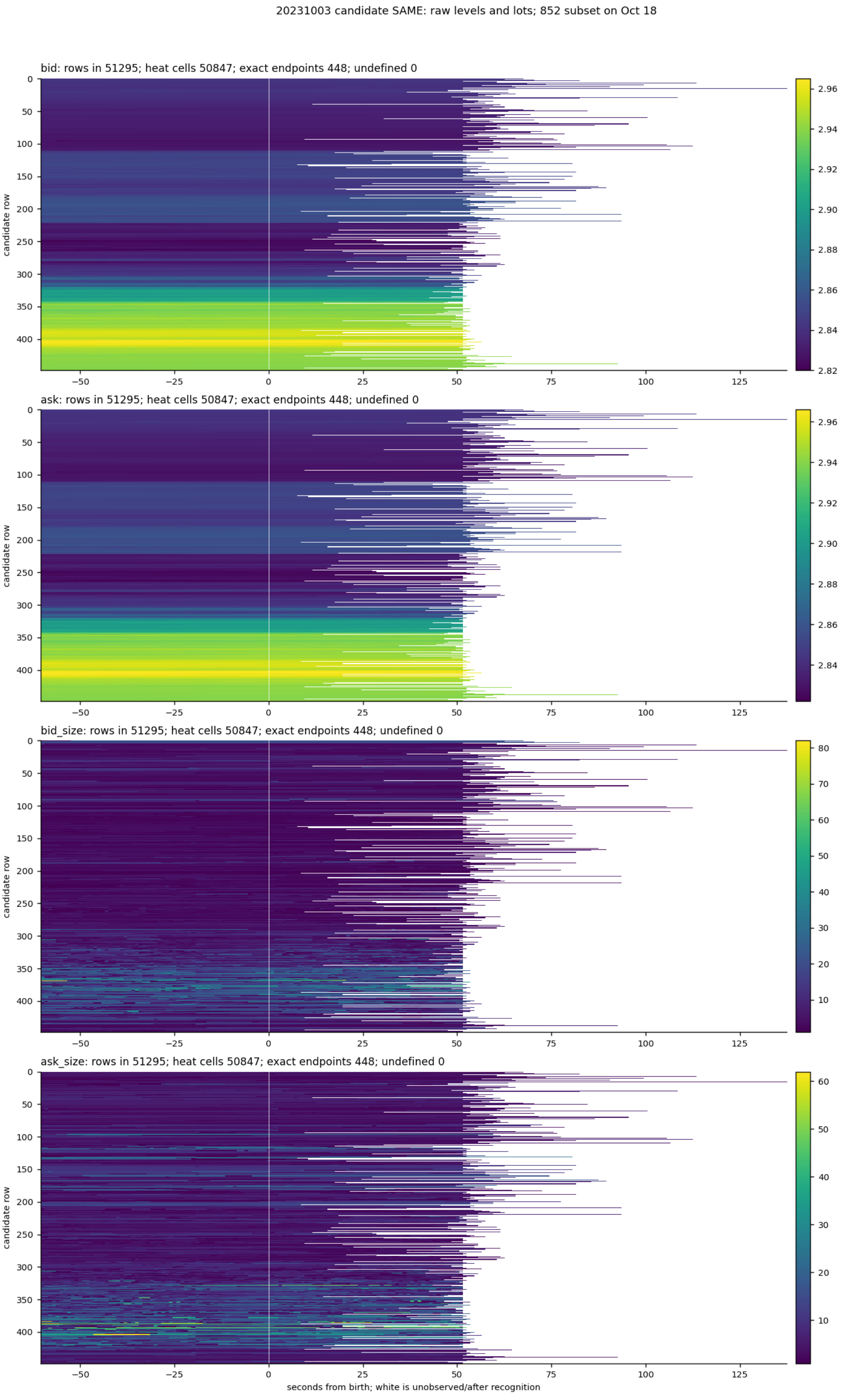


20231003-candidate-SAME-endpoints.png

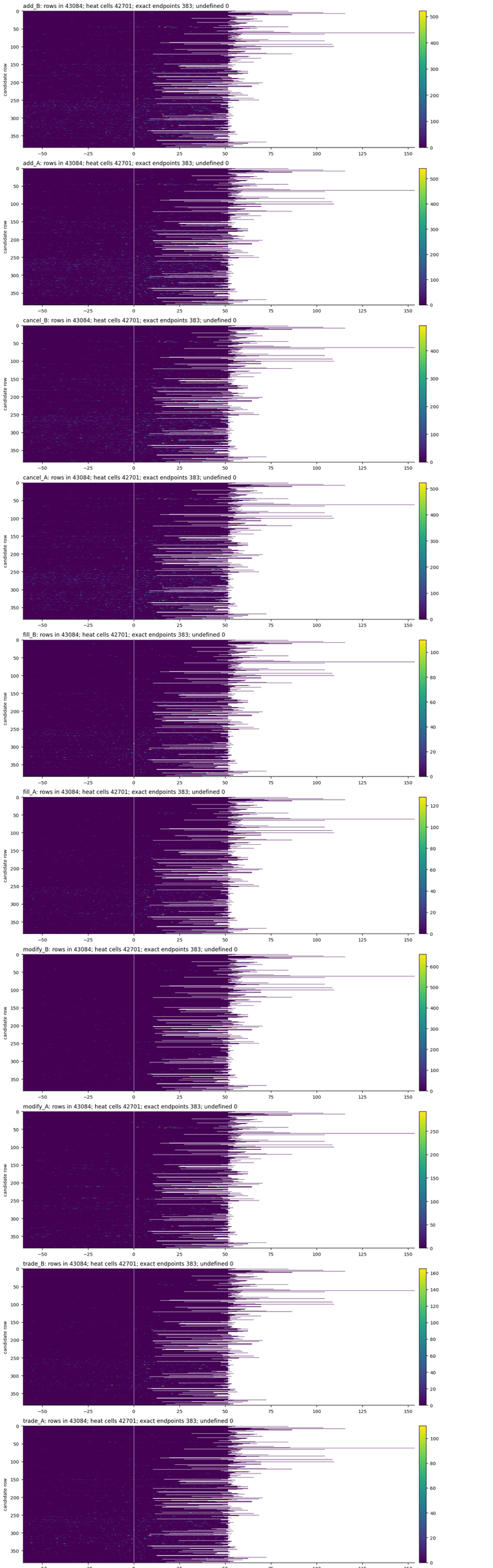
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231003 candidate SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 51295; heat cells 33433; exact endpoints 258; undefined 17604
candidate row
0
50
100
150
200
250
300
350
400
−50
−25
0
25
50
75
100
125
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

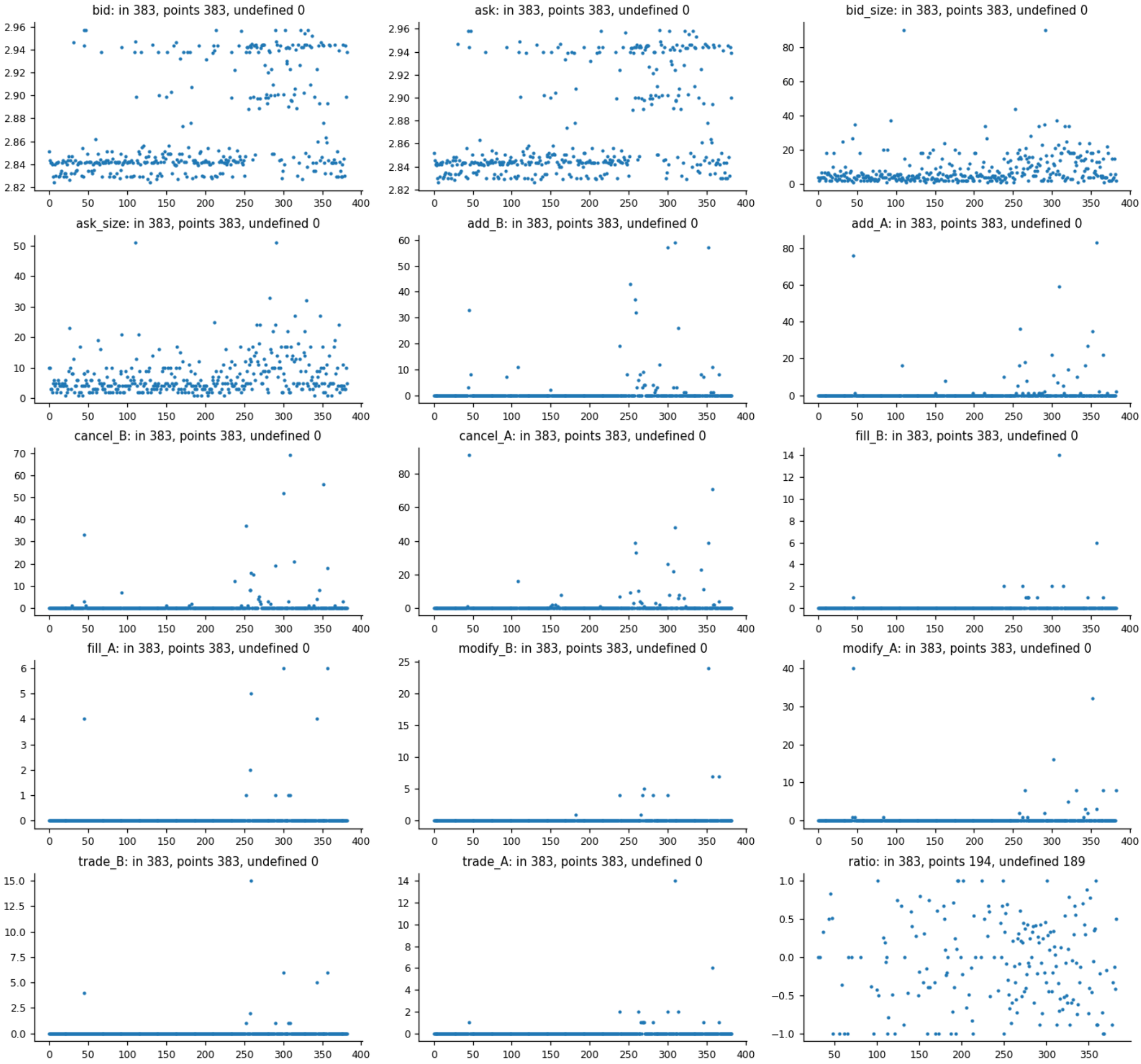

20231003 candidate SAME: raw levels and lots; 852 subset on Oct 18
bid: rows in 51295; heat cells 50847; exact endpoints 448; undefined 0
ask: rows in 51295; heat cells 50847; exact endpoints 448; undefined 0
bid_size: rows in 51295; heat cells 50847; exact endpoints 448; undefined 0
ask_size: rows in 51295; heat cells 50847; exact endpoints 448; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

20231003 control1 FLIP: raw levels and lots; 852 subset on Oct 18



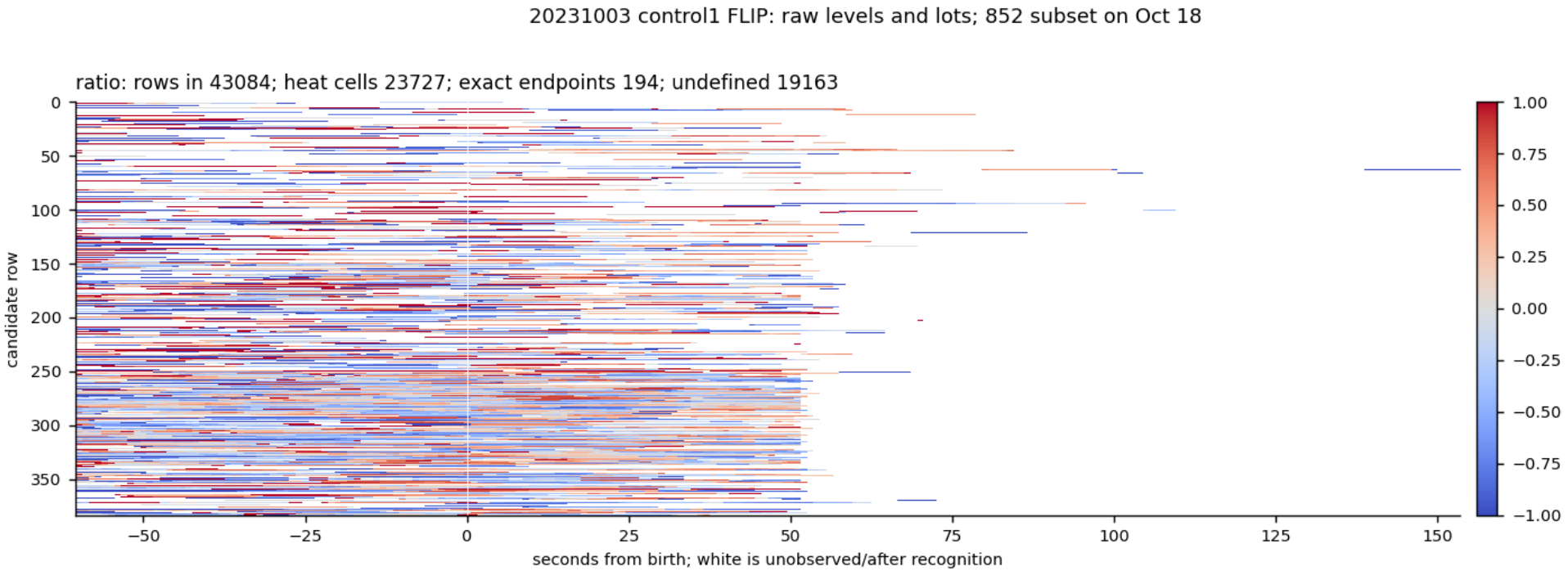
20231003-control1-FLIP-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



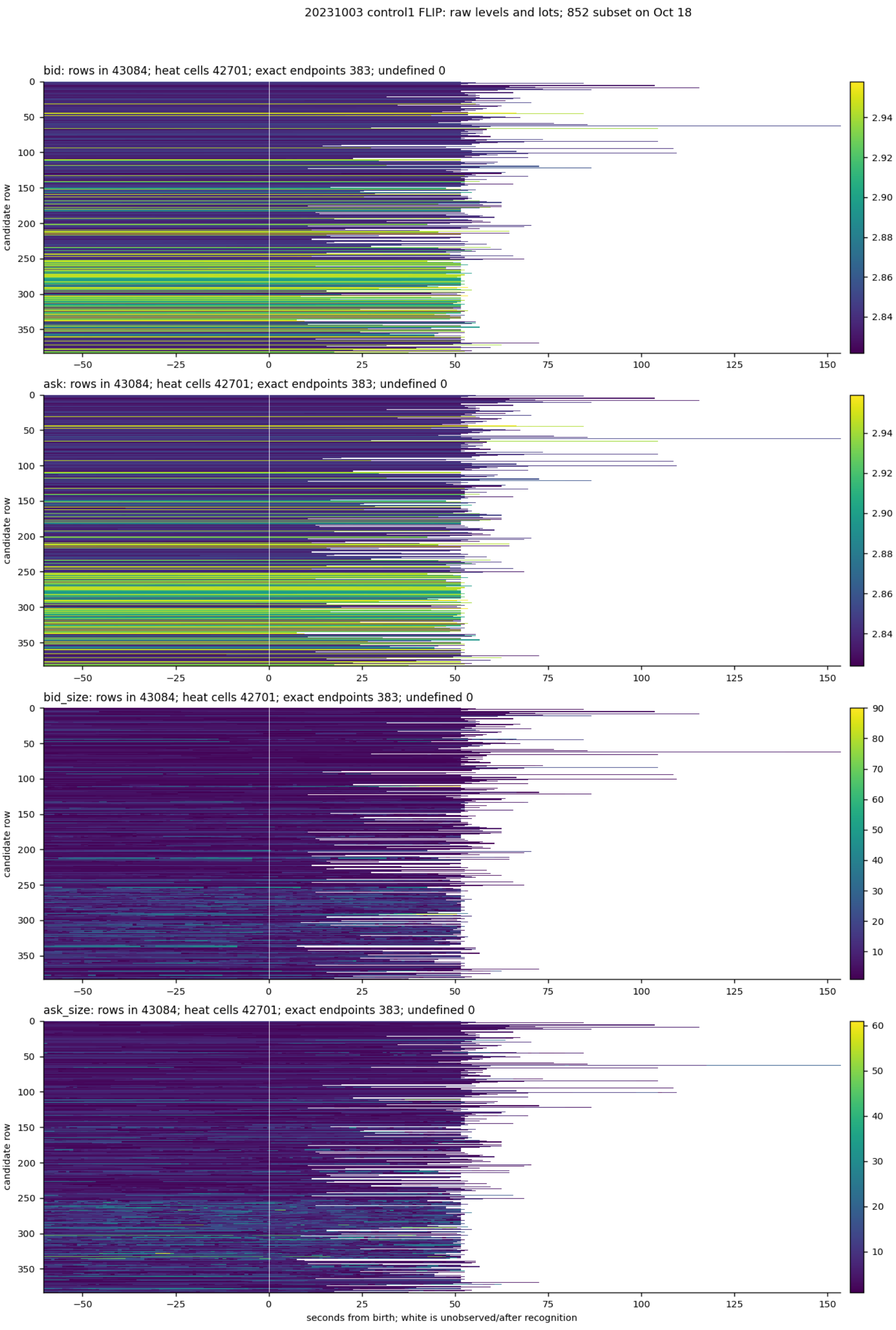
20231003-control1-FLIP-flow.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



20231003-control1-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

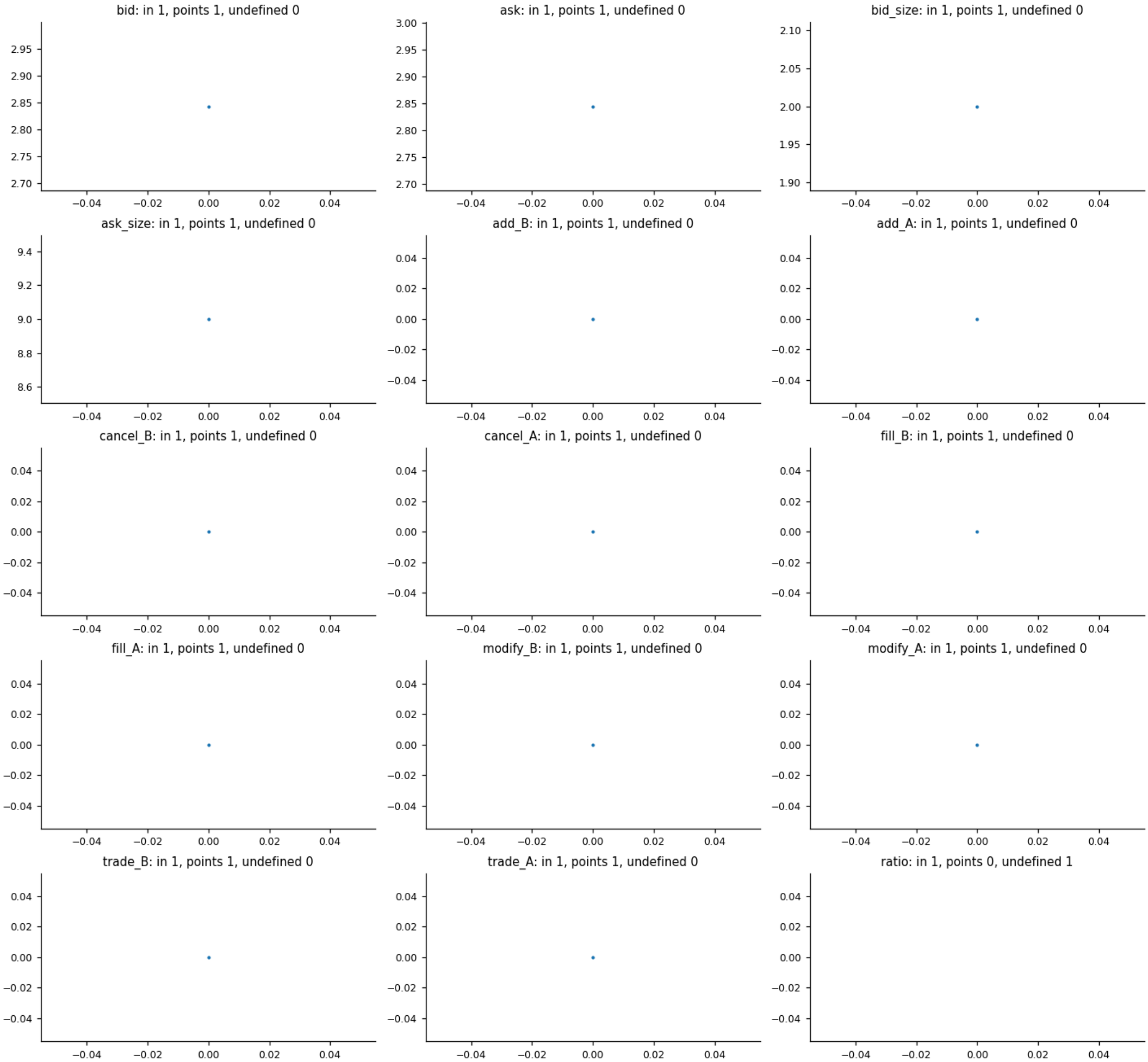


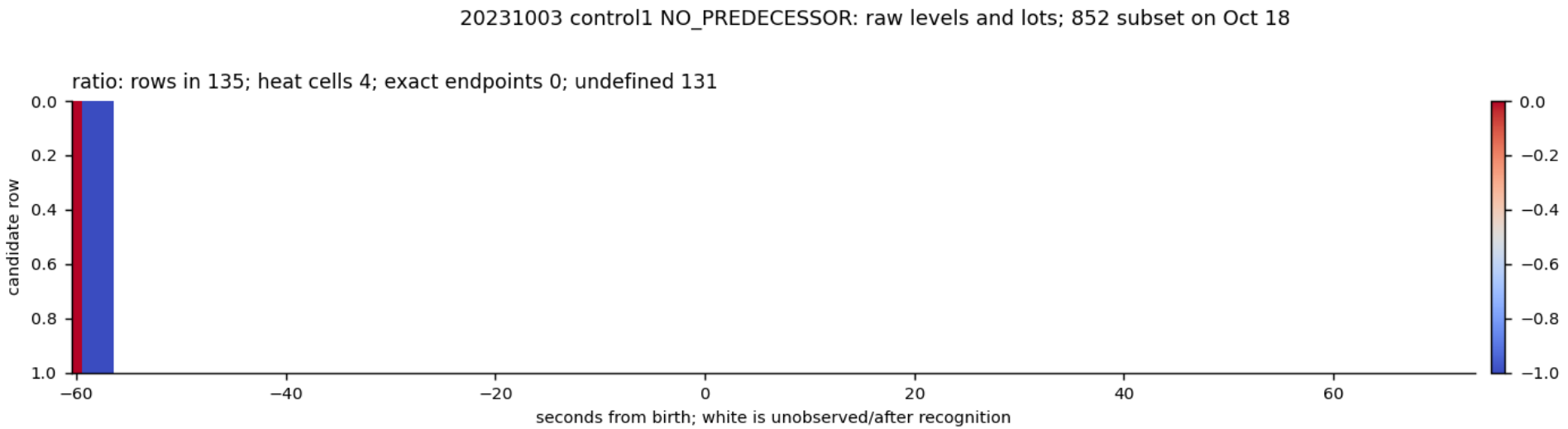
20231003-control1-NO_PREDECESSOR-actions.png

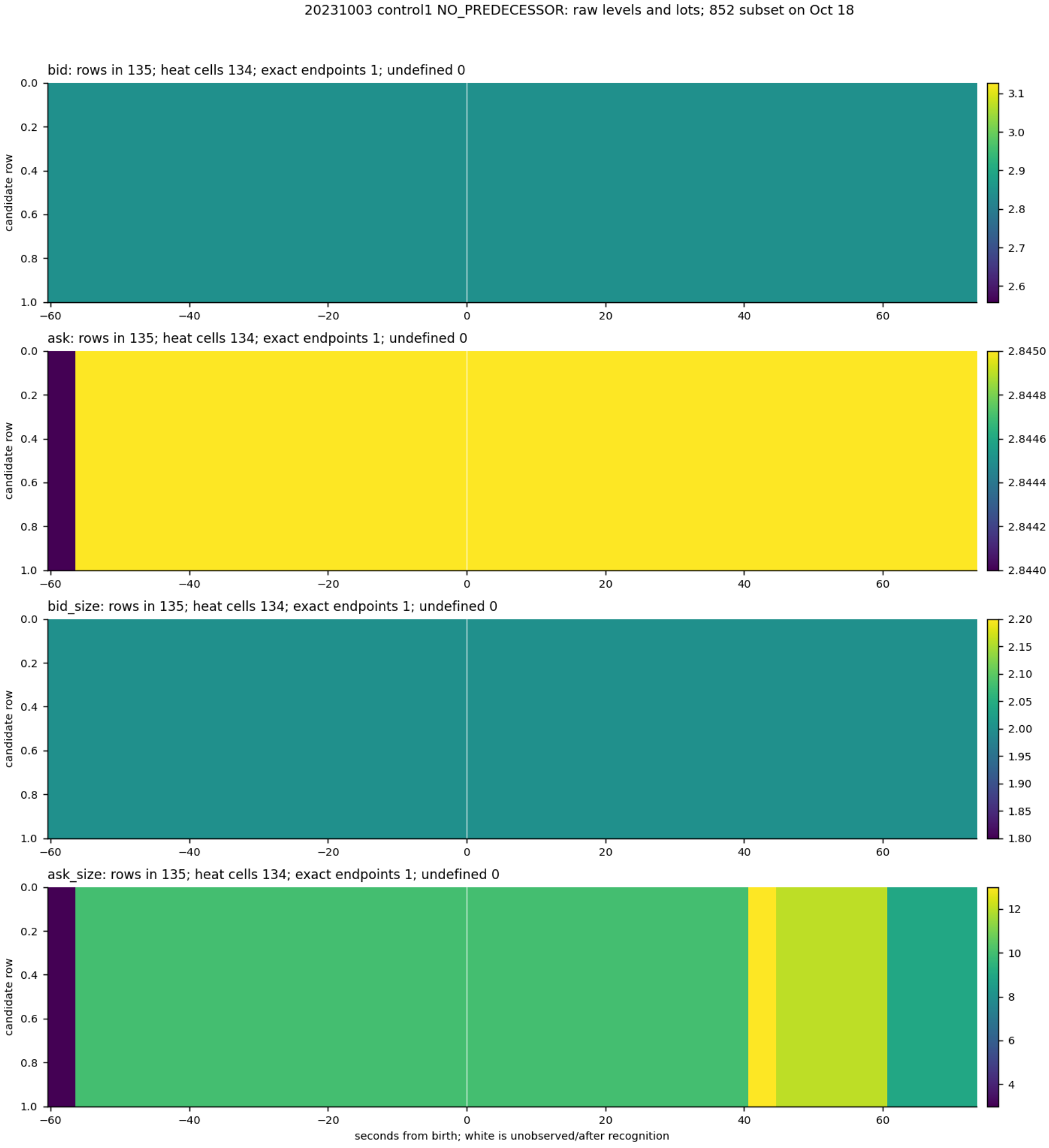
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

20231003-control1-NO_PREDECESSOR-endpoints.png

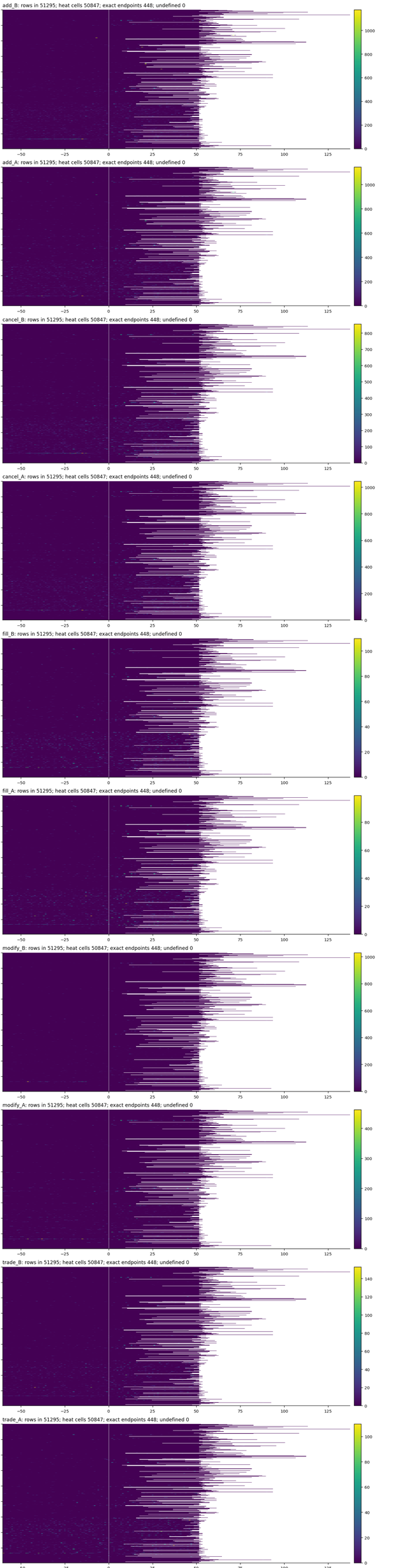
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231003 control1 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
ratio: rows in 135; heat cells 4; exact endpoints 0; undefined 131
candidate row
0.0
0.2
0.4
0.6
0.8
1.0
−60
−40
−20
0
20
40
60
seconds from birth; white is unobserved/after recognition
0.0
−0.2
−0.4
−0.6
−0.8
−1.0

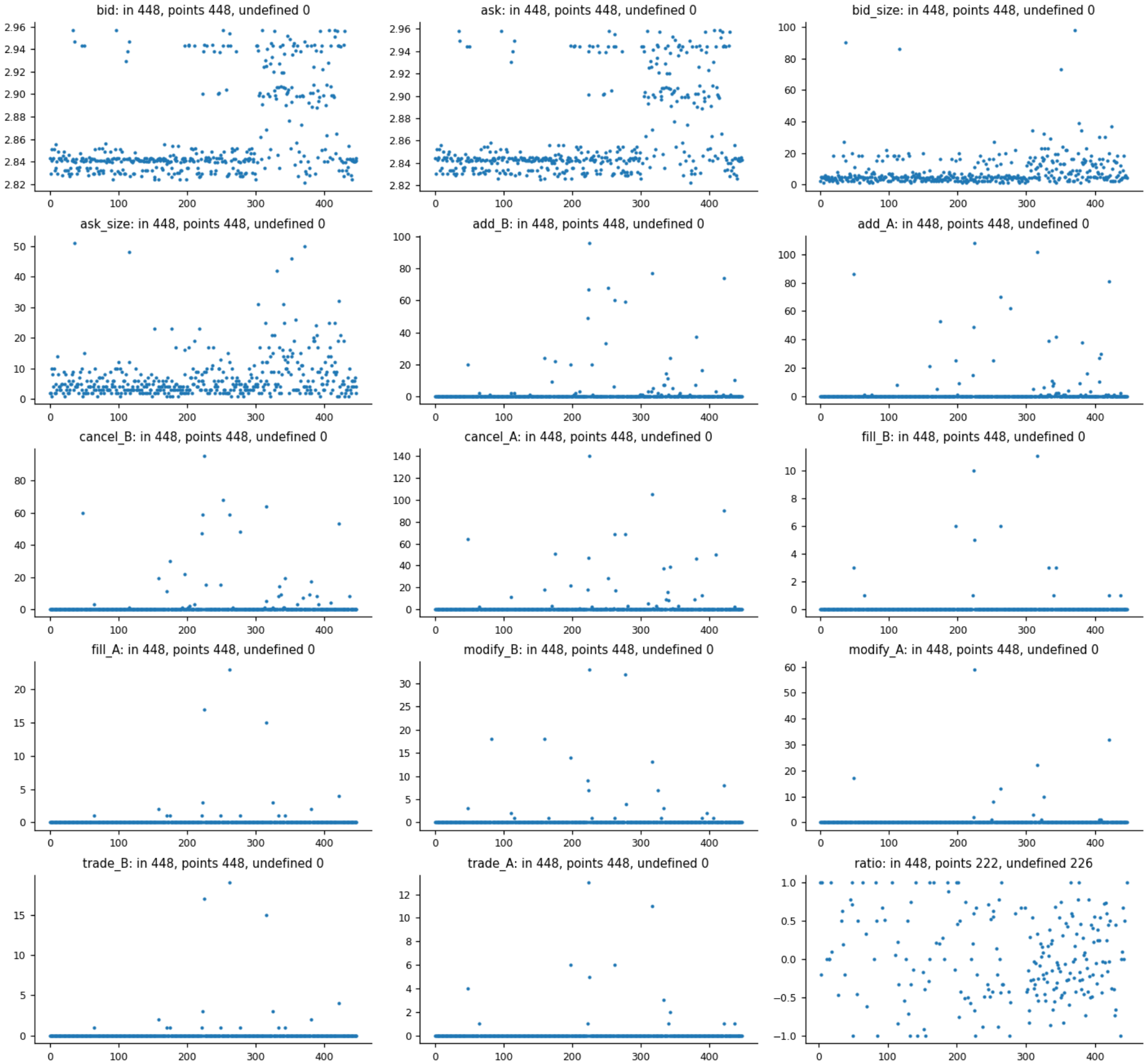

20231003 control1 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 135; heat cells 134; exact endpoints 1; undefined 0
ask: rows in 135; heat cells 134; exact endpoints 1; undefined 0
bid_size: rows in 135; heat cells 134; exact endpoints 1; undefined 0
ask_size: rows in 135; heat cells 134; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

20231003 control1 SAME: raw levels and lots; 852 subset on Oct 18

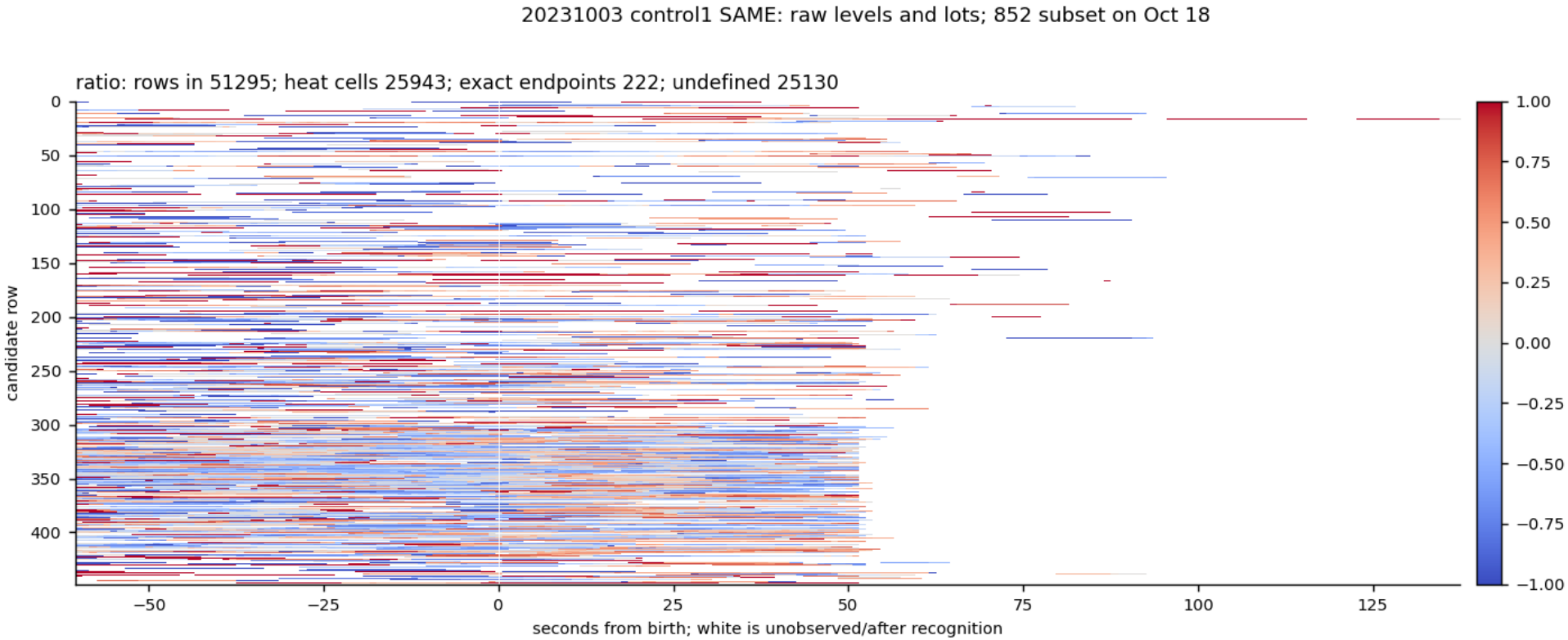


20231003-control1-SAME-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

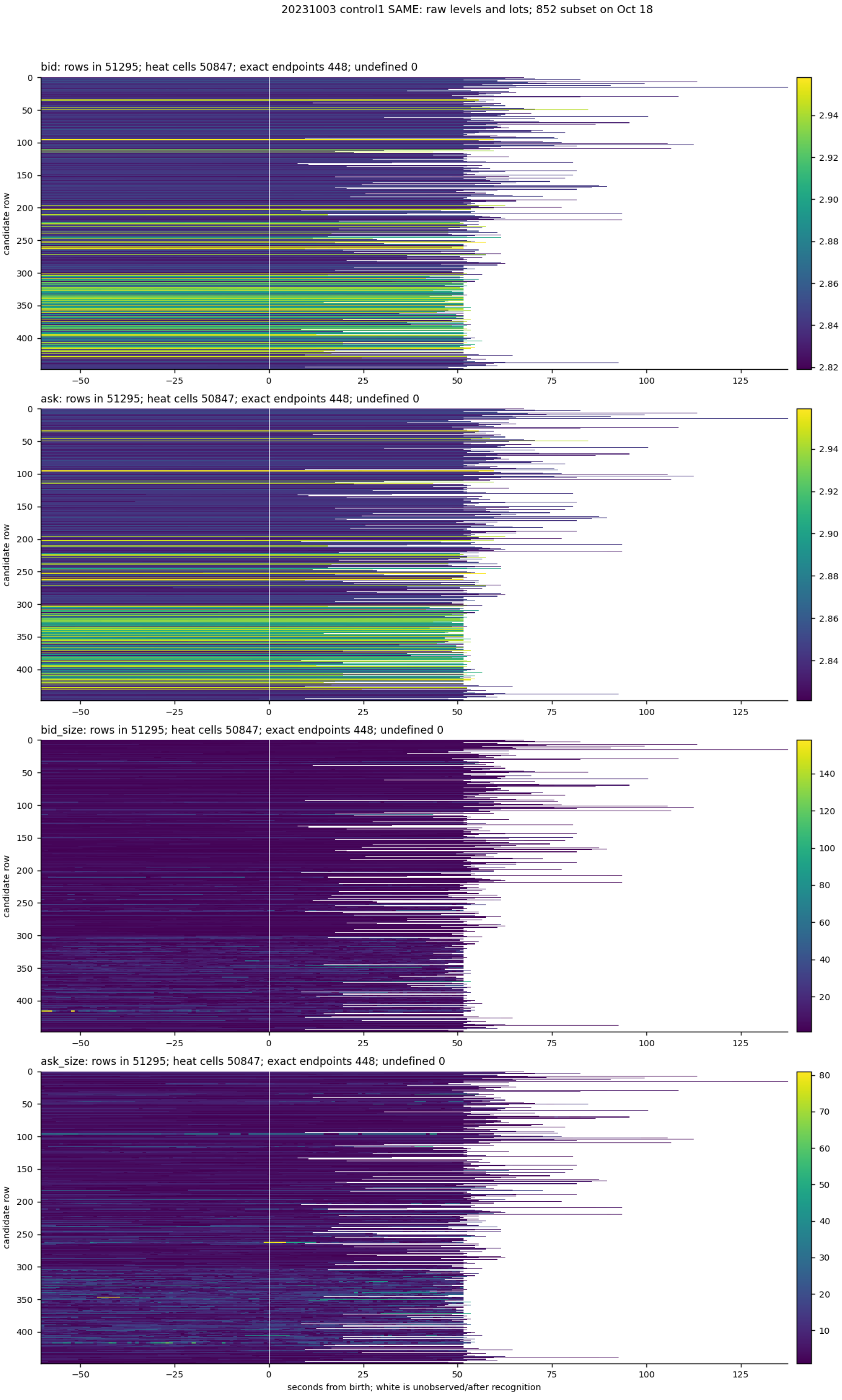


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

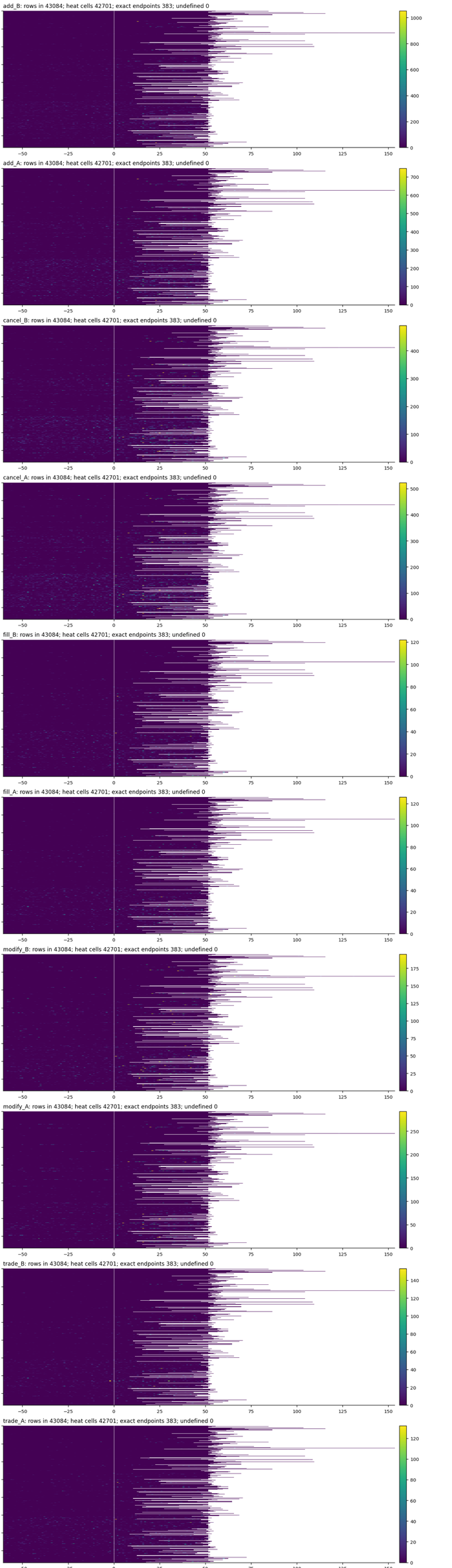


20231003-control1-SAME-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

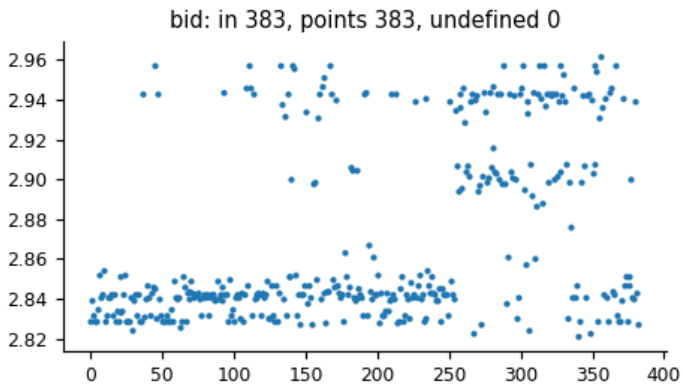


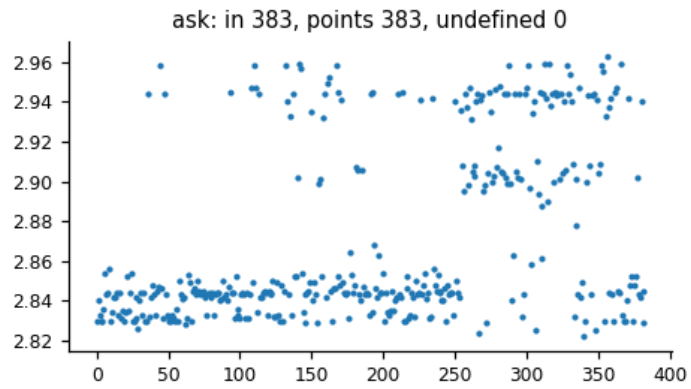
20231003 control2 FLIP: raw levels and lots; 852 subset on Oct 18

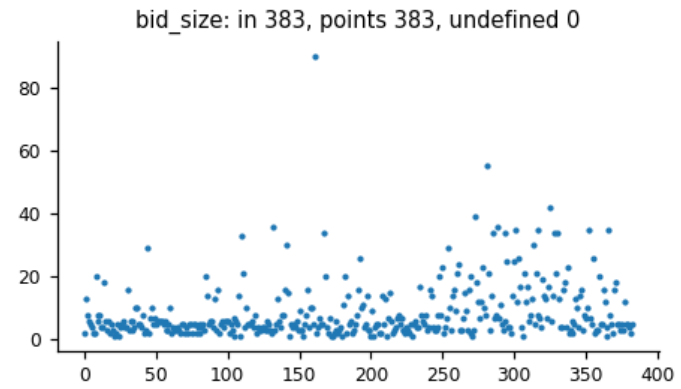


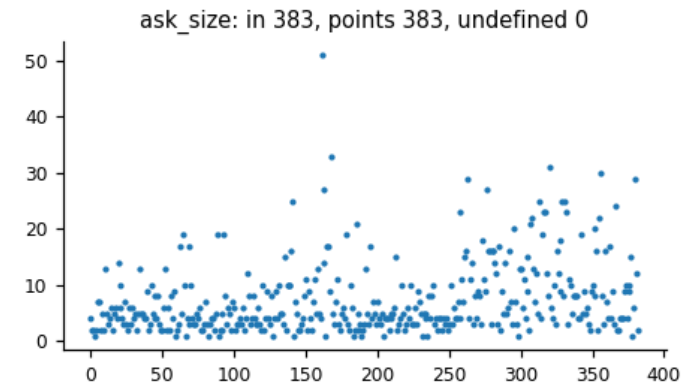
20231003-control2-FLIP-endpoints.png

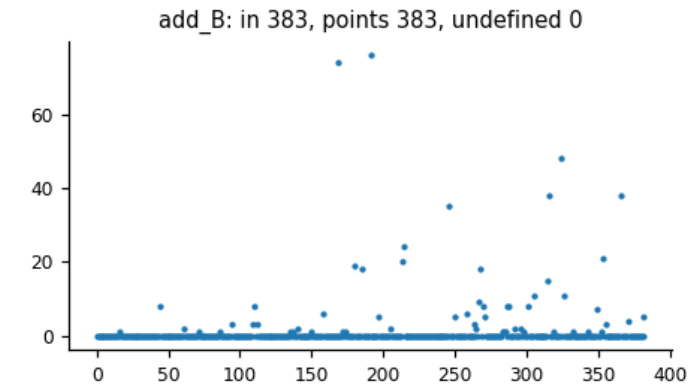
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

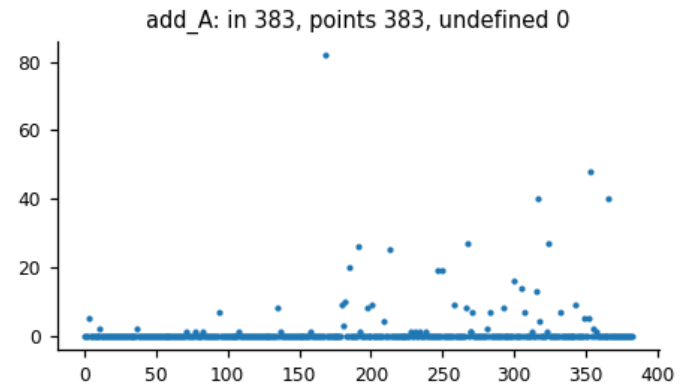


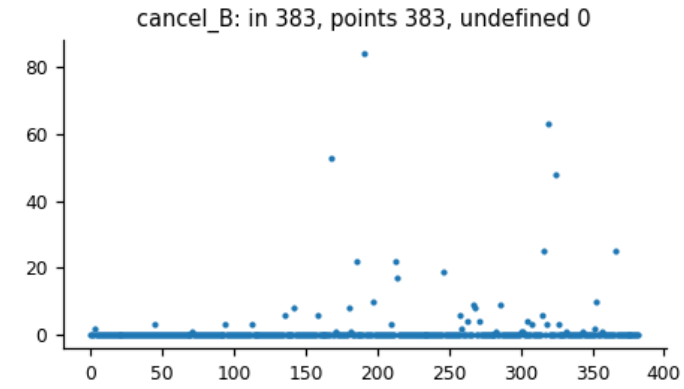


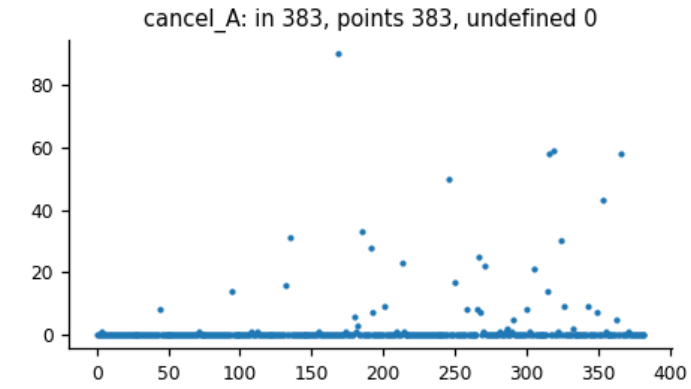


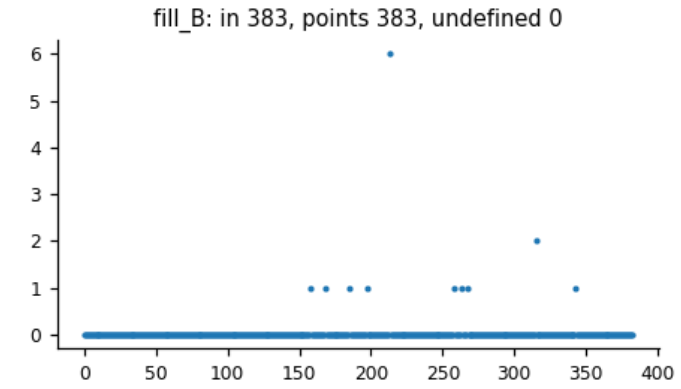


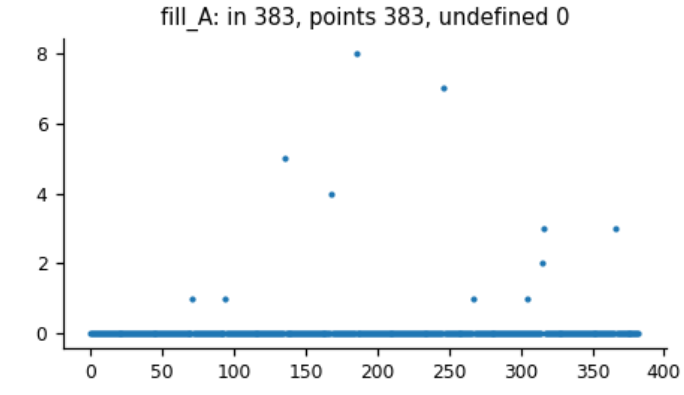


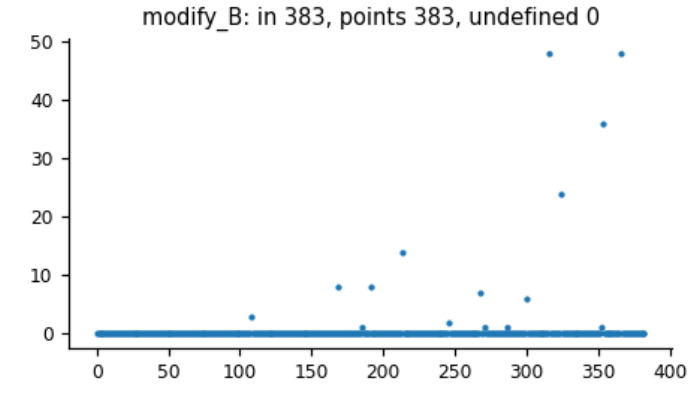


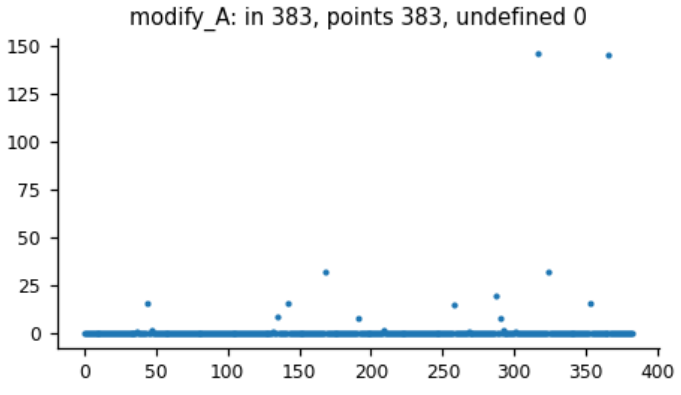


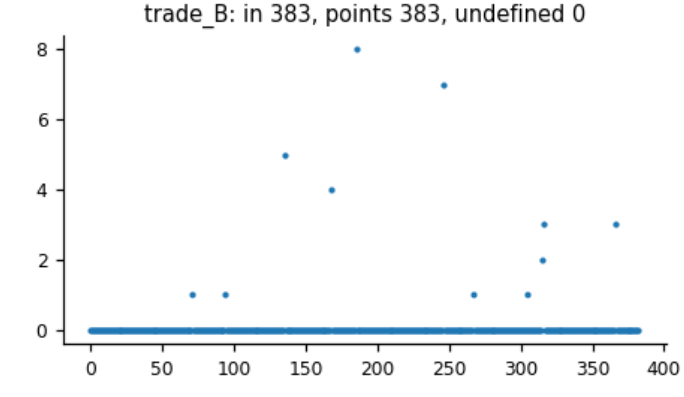


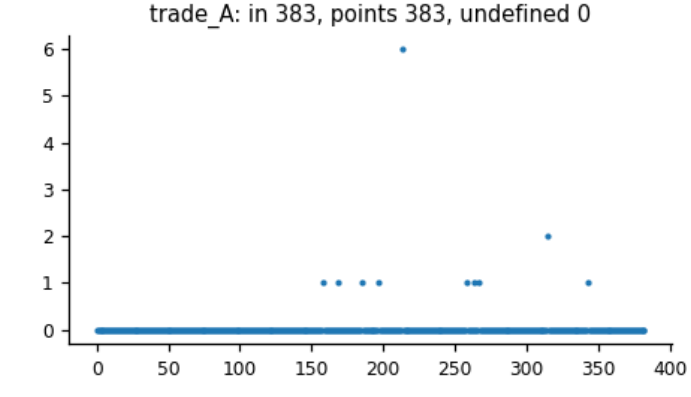


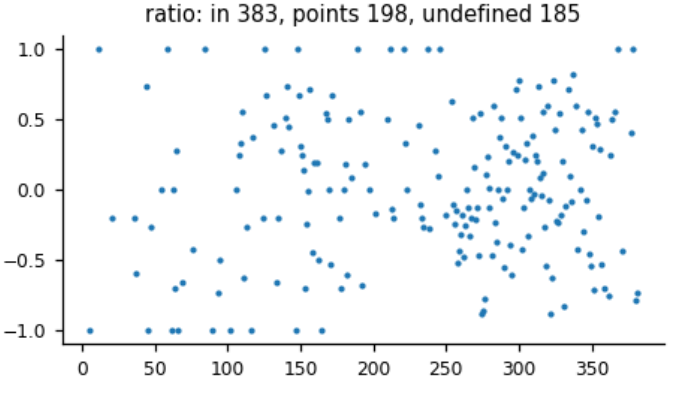


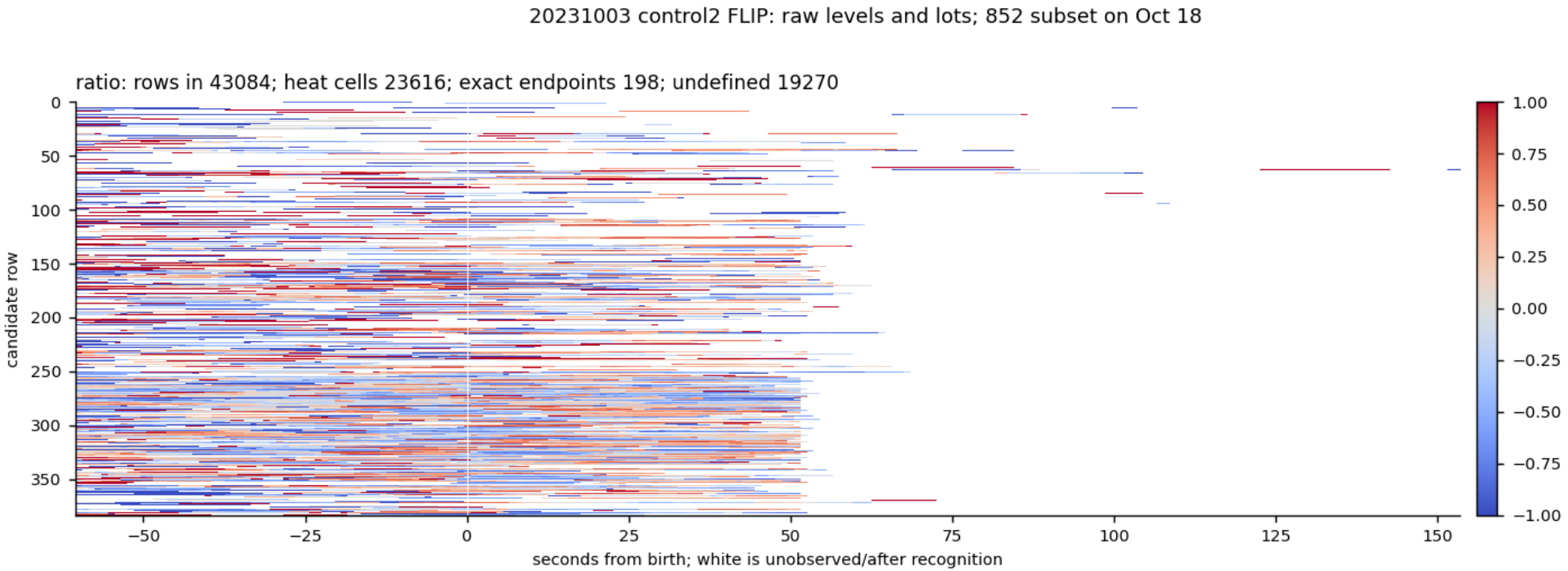






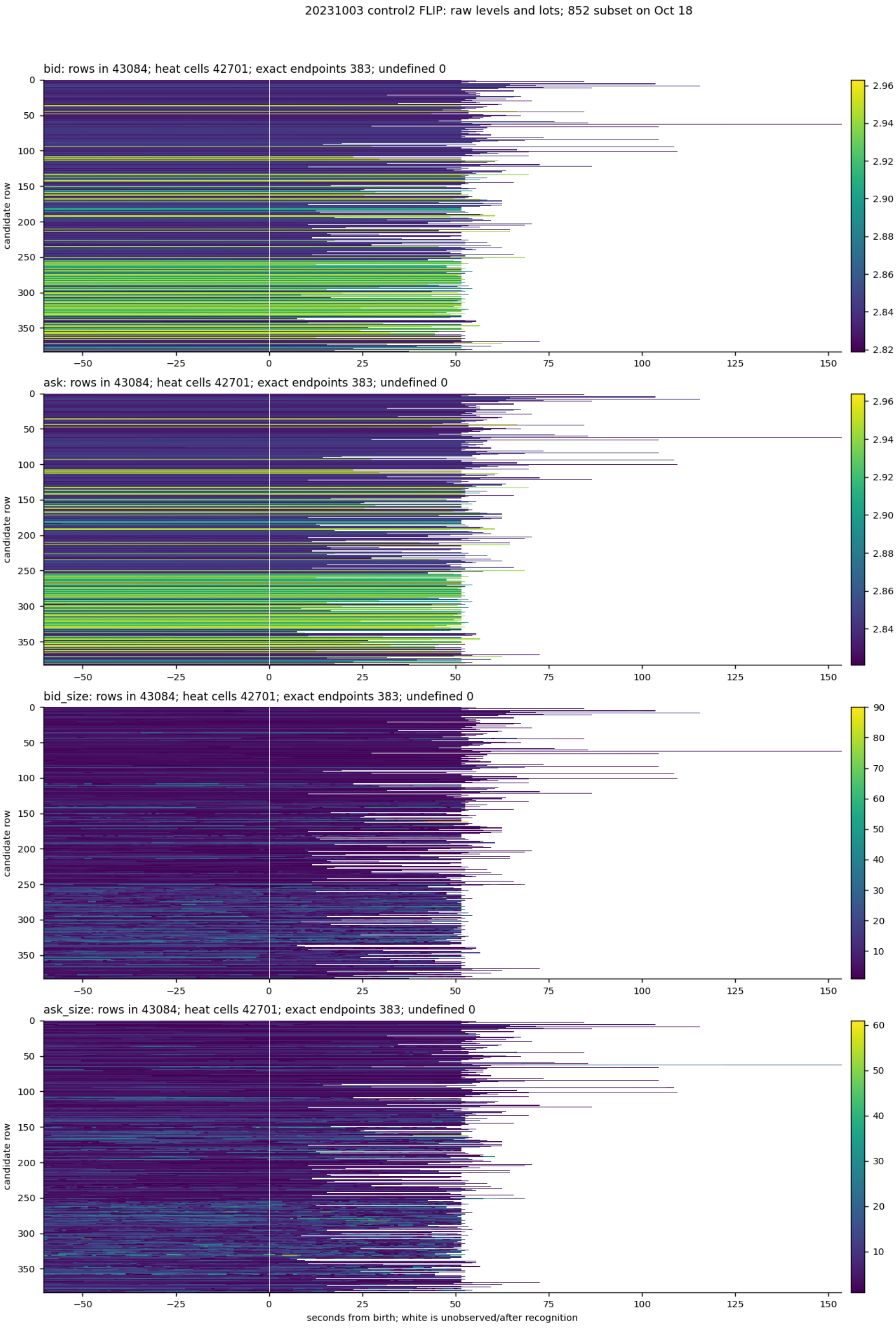





20231003 control2 FLIP: raw levels and lots; 852 subset on Oct 18
ratio: rows in 43084; heat cells 23616; exact endpoints 198; undefined 19270
candidate row
0
50
100
150
200
250
300
350
−50
−25
0
25
50
75
100
125
150
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

20231003-control2-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

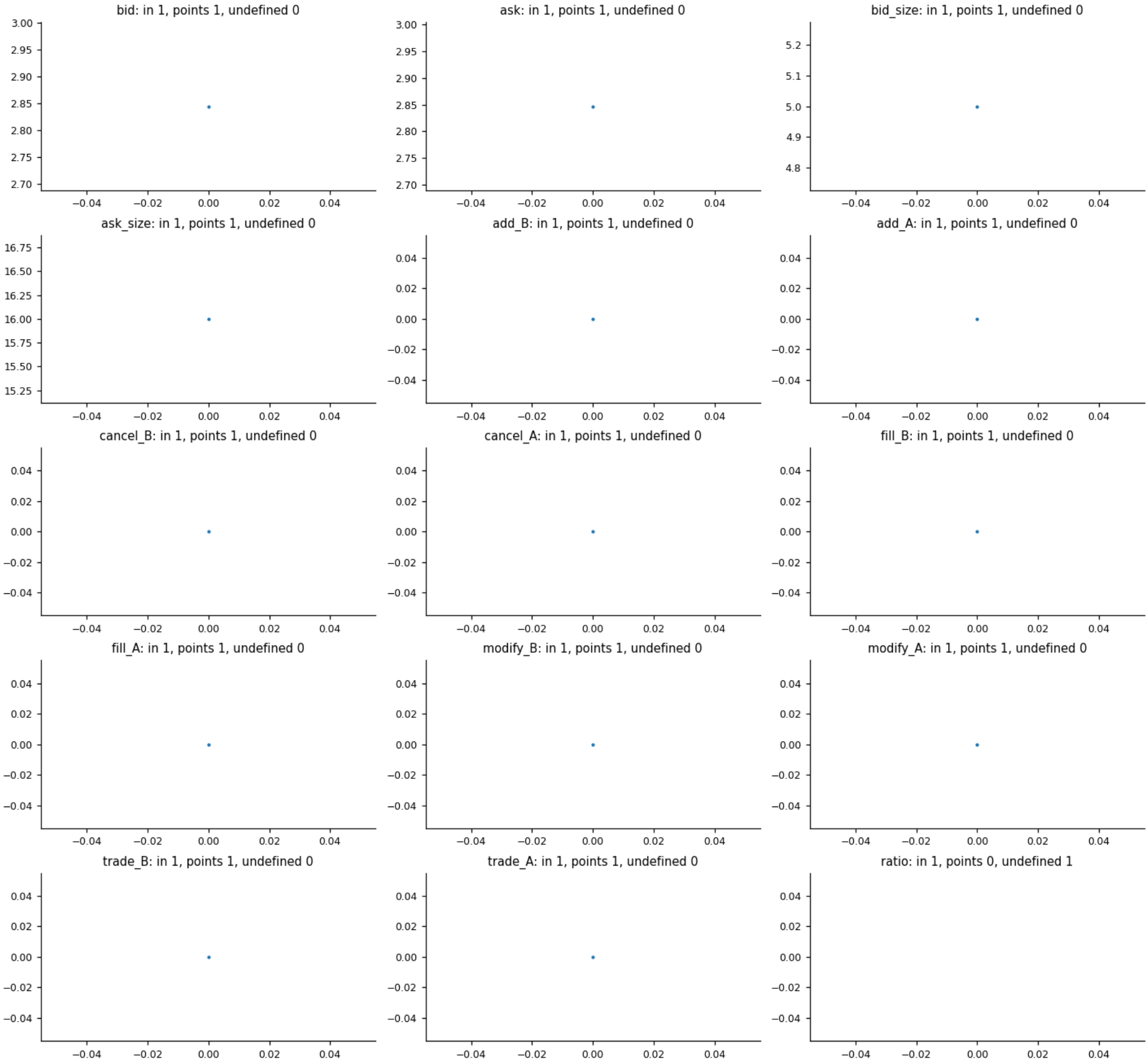


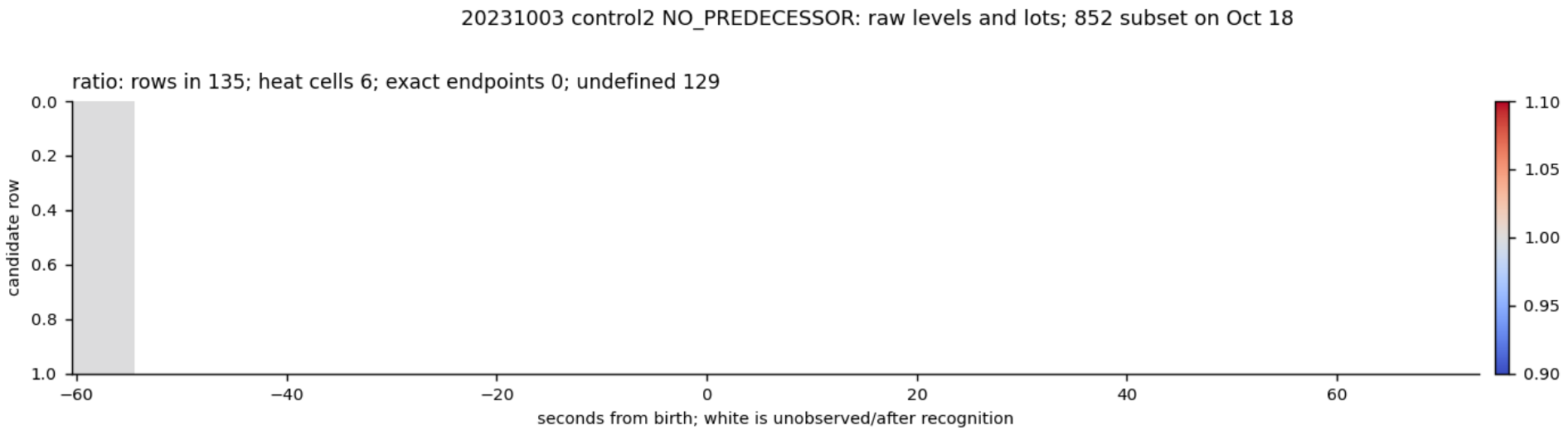
20231003-control2-NO_PREDECESSOR-actions.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

20231003-control2-NO_PREDECESSOR-endpoints.png

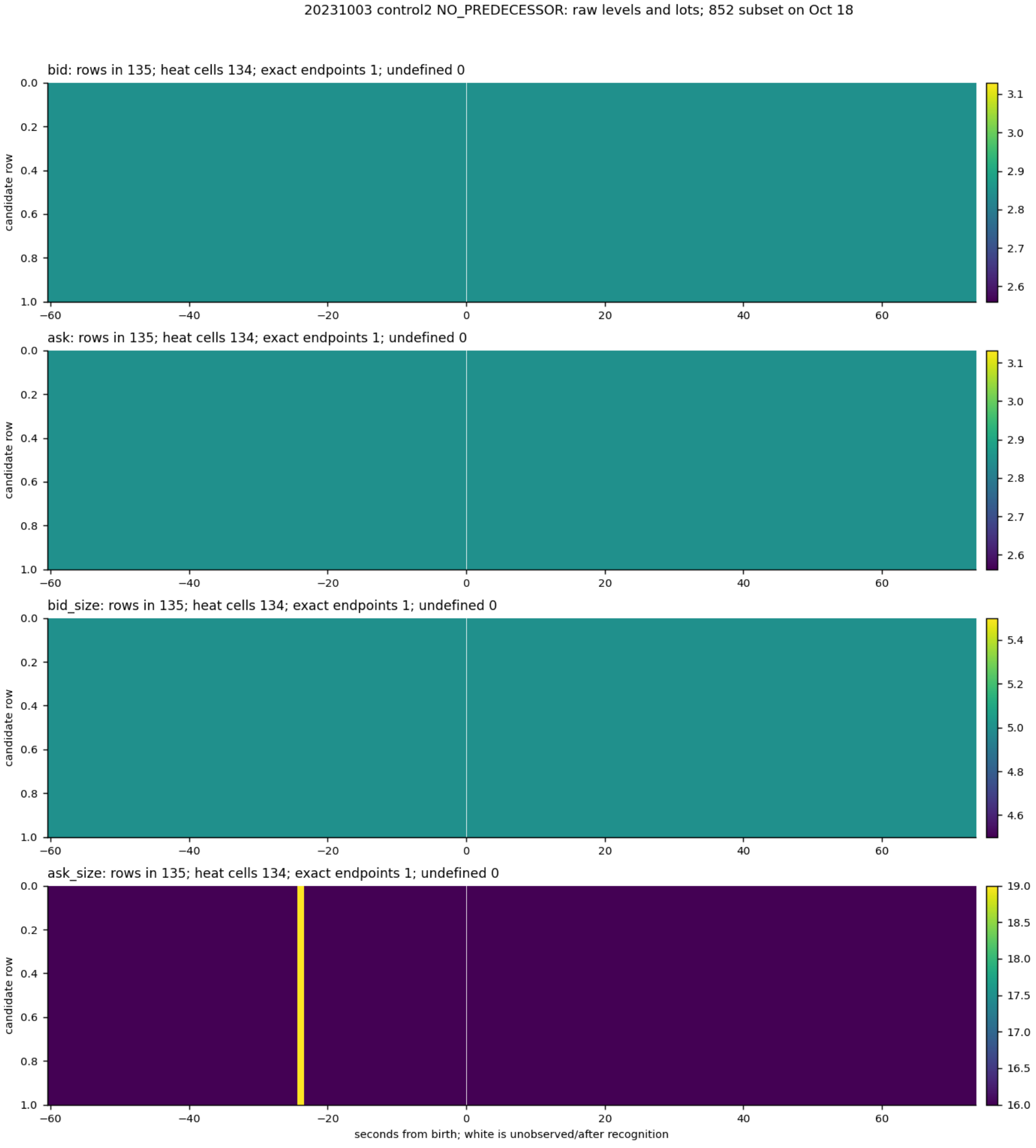
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



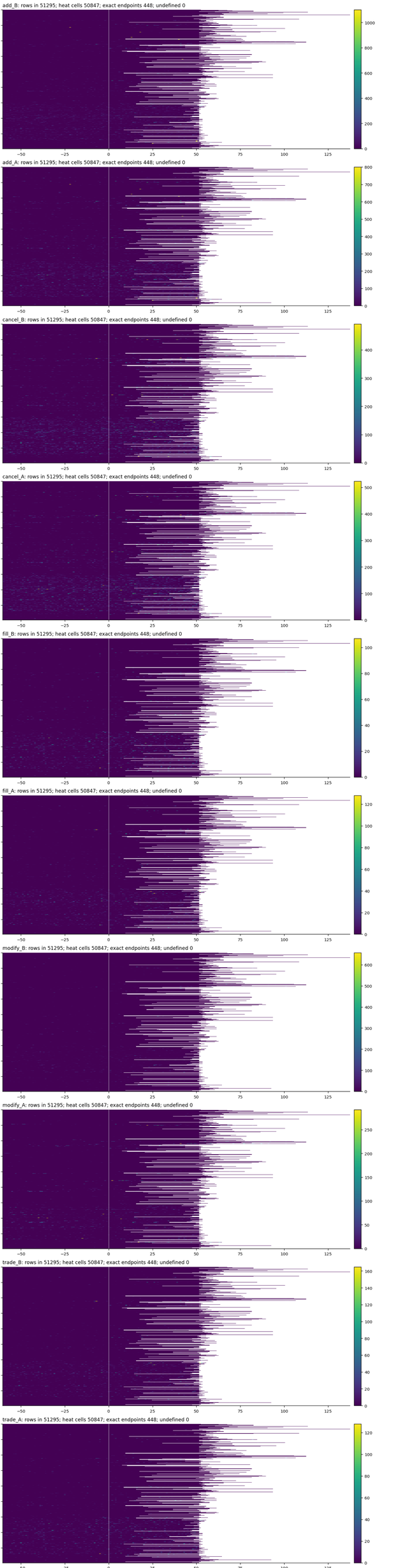

20231003 control2 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
ratio: rows in 135; heat cells 6; exact endpoints 0; undefined 129
candidate row
0.0
0.2
0.4
0.6
0.8
1.0
−60
−40
−20
0
20
40
60
seconds from birth; white is unobserved/after recognition
1.10
1.05
1.00
0.95
0.90

20231003-control2-NO_PREDECESSOR-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



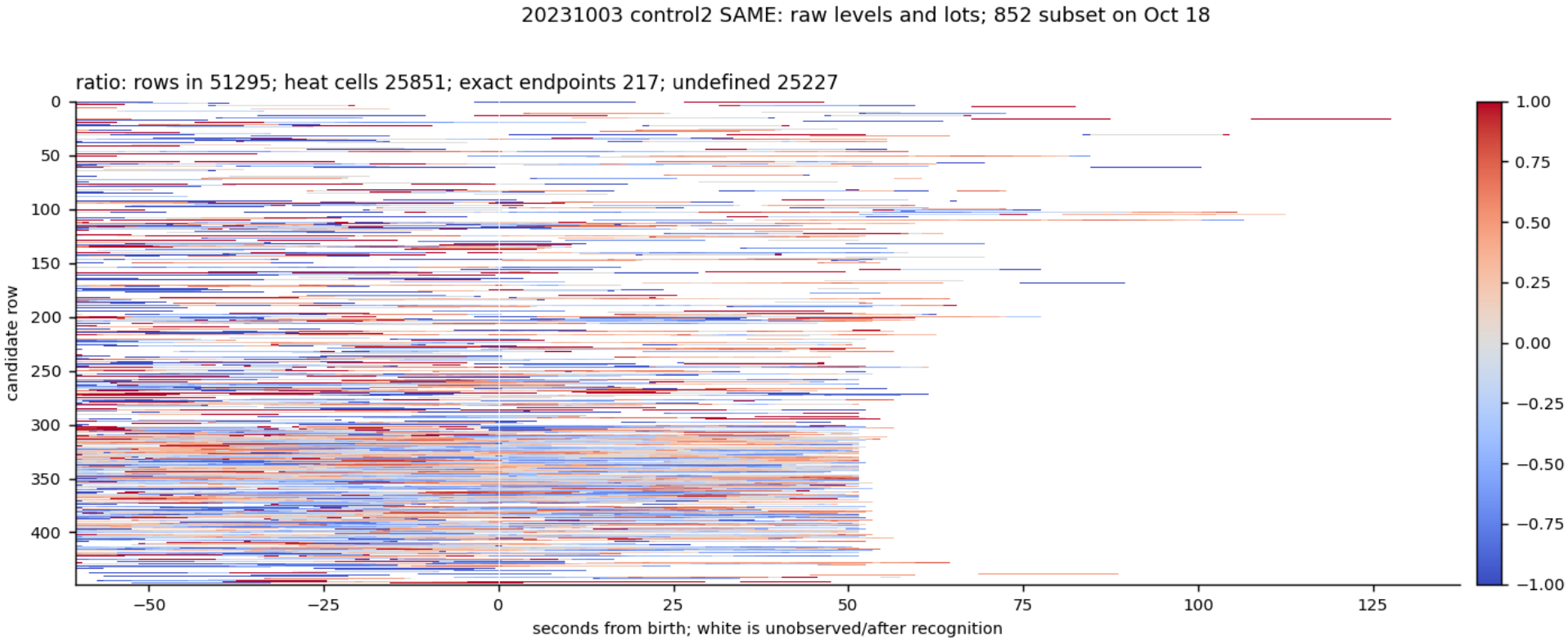
20231003 control2 SAME: raw levels and lots; 852 subset on Oct 18



20231003-control2-SAME-endpoints.png

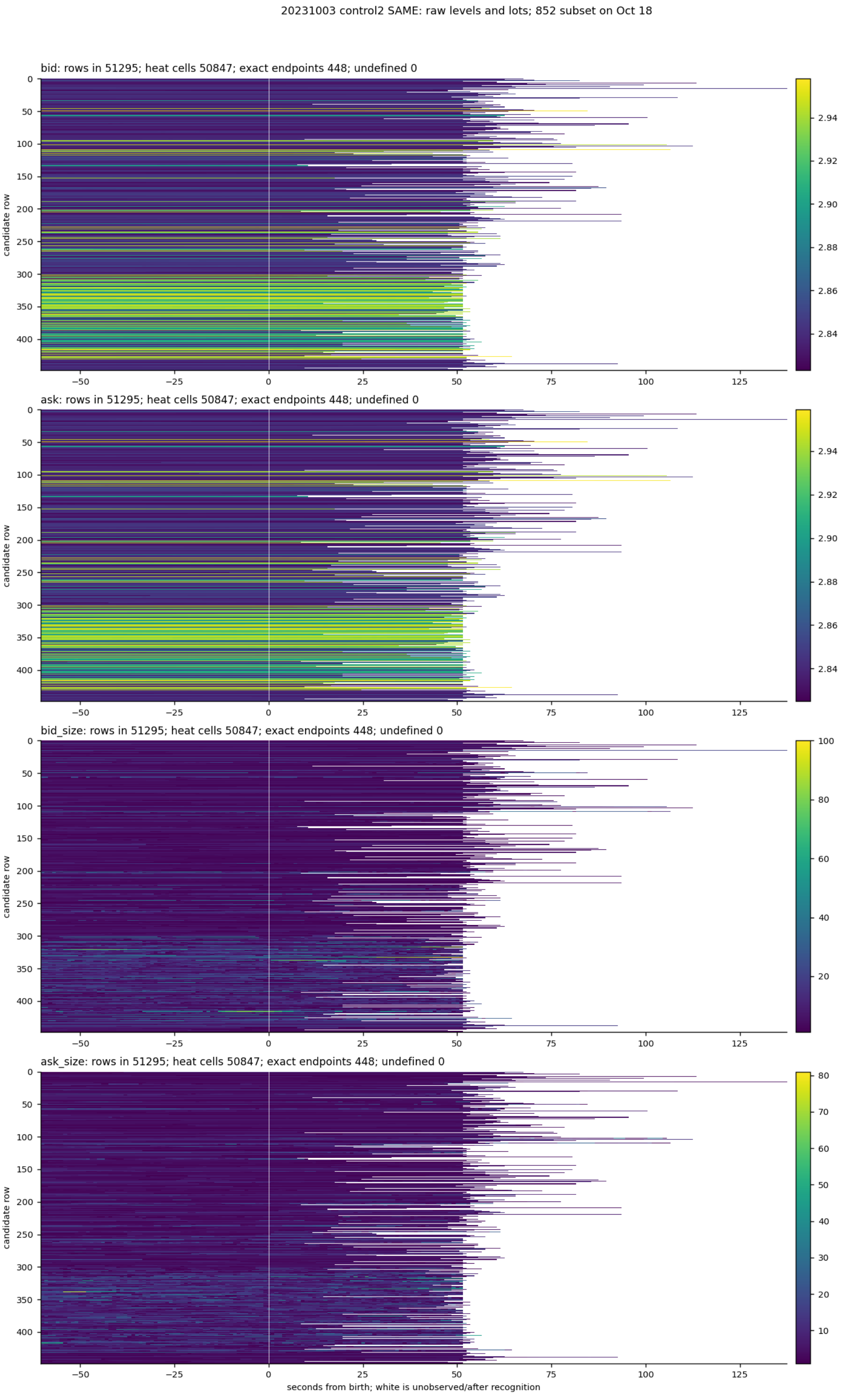
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231003 control2 SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 51295; heat cells 25851; exact endpoints 217; undefined 25227
candidate row
0
50
100
150
200
250
300
350
400
−50
−25
0
25
50
75
100
125
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

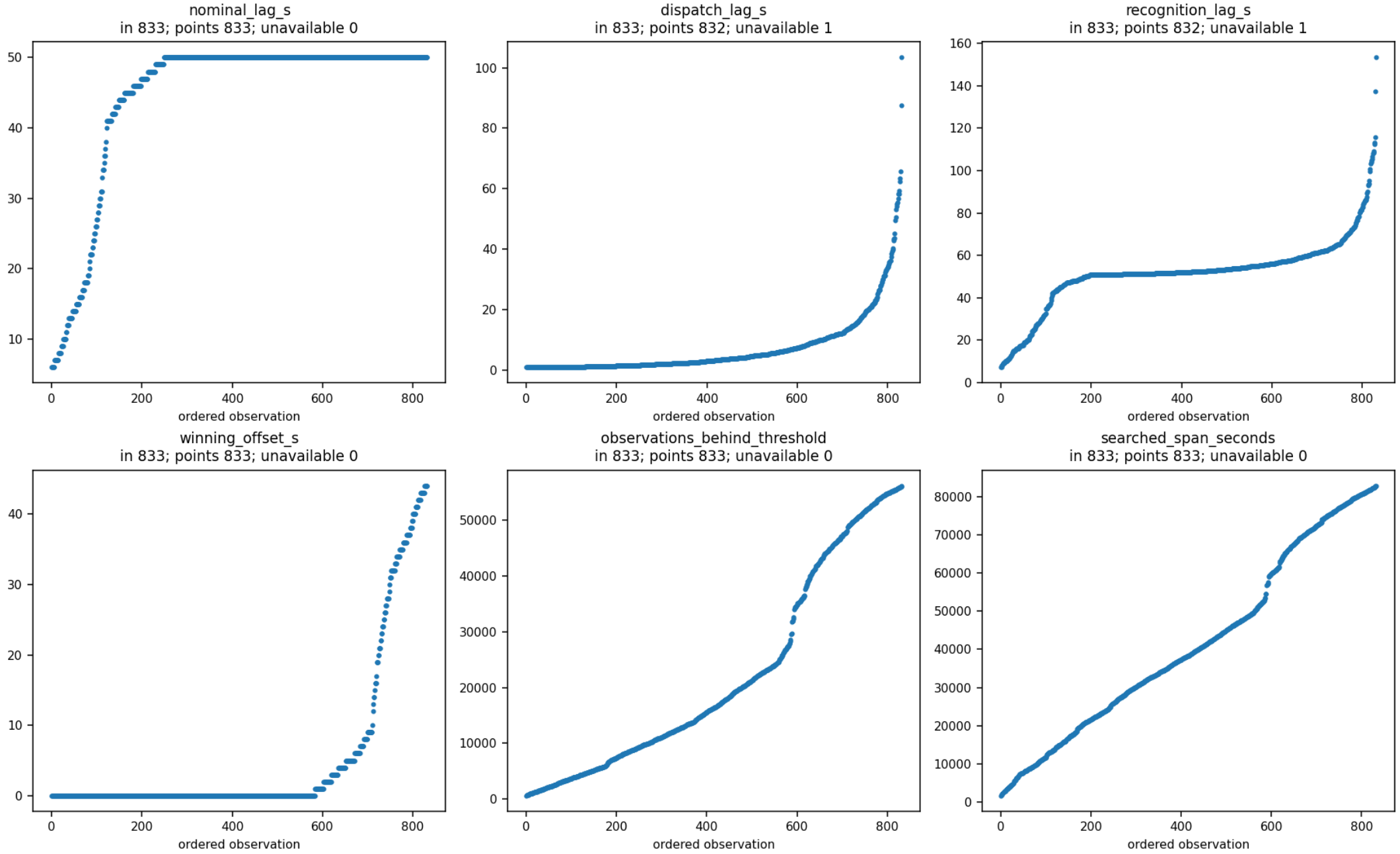
20231003-control2-SAME-quotes.png

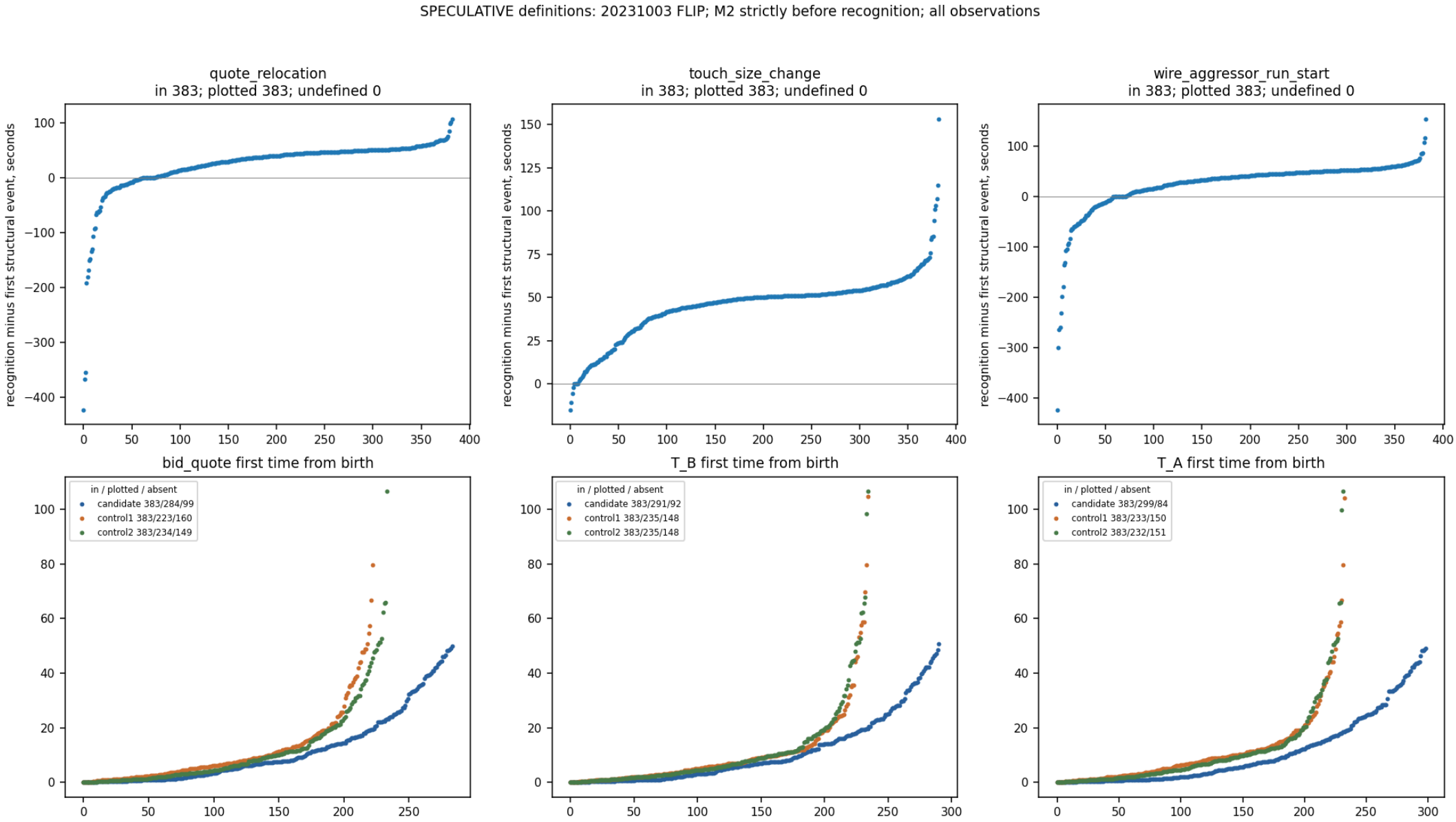
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

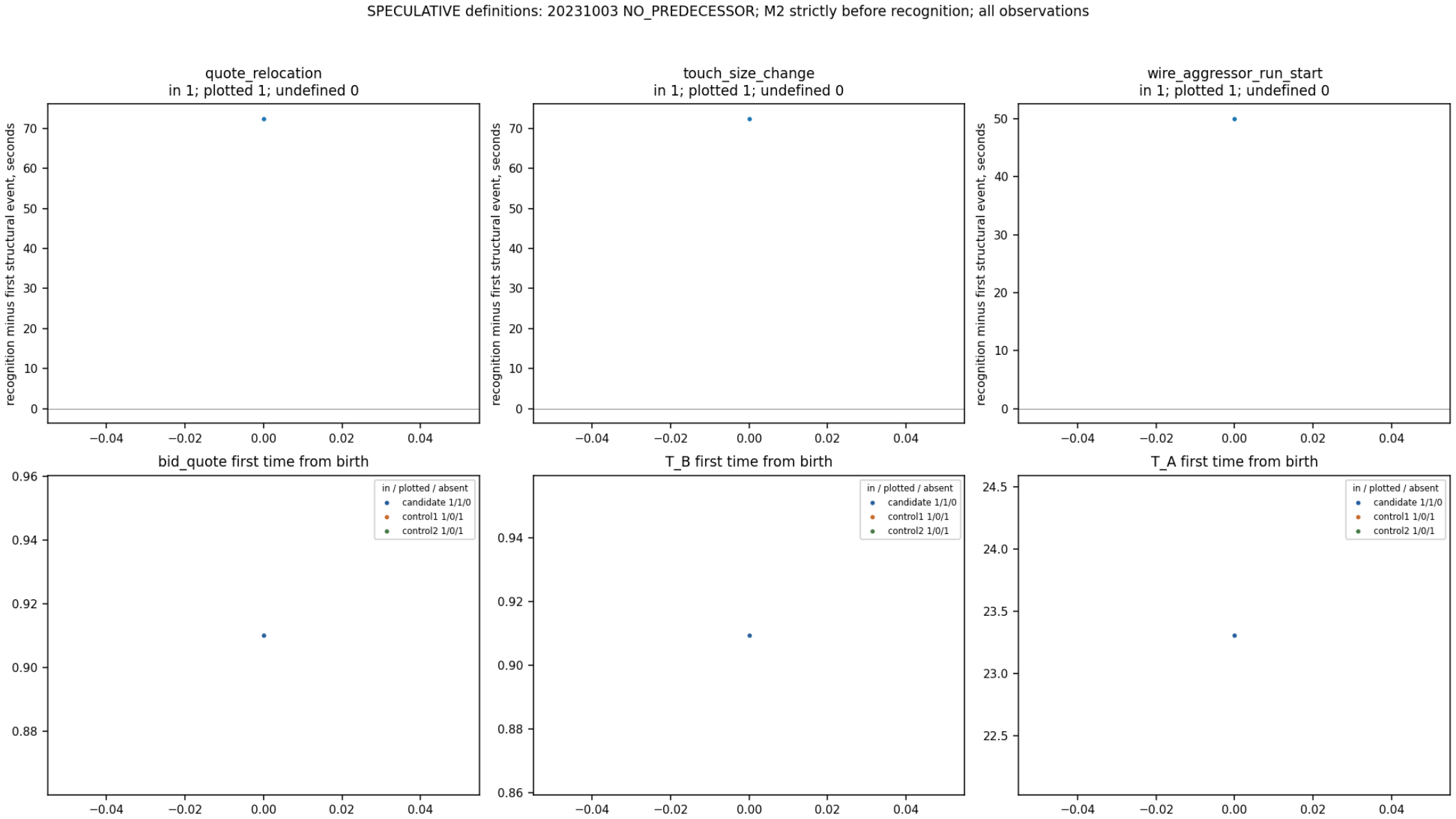


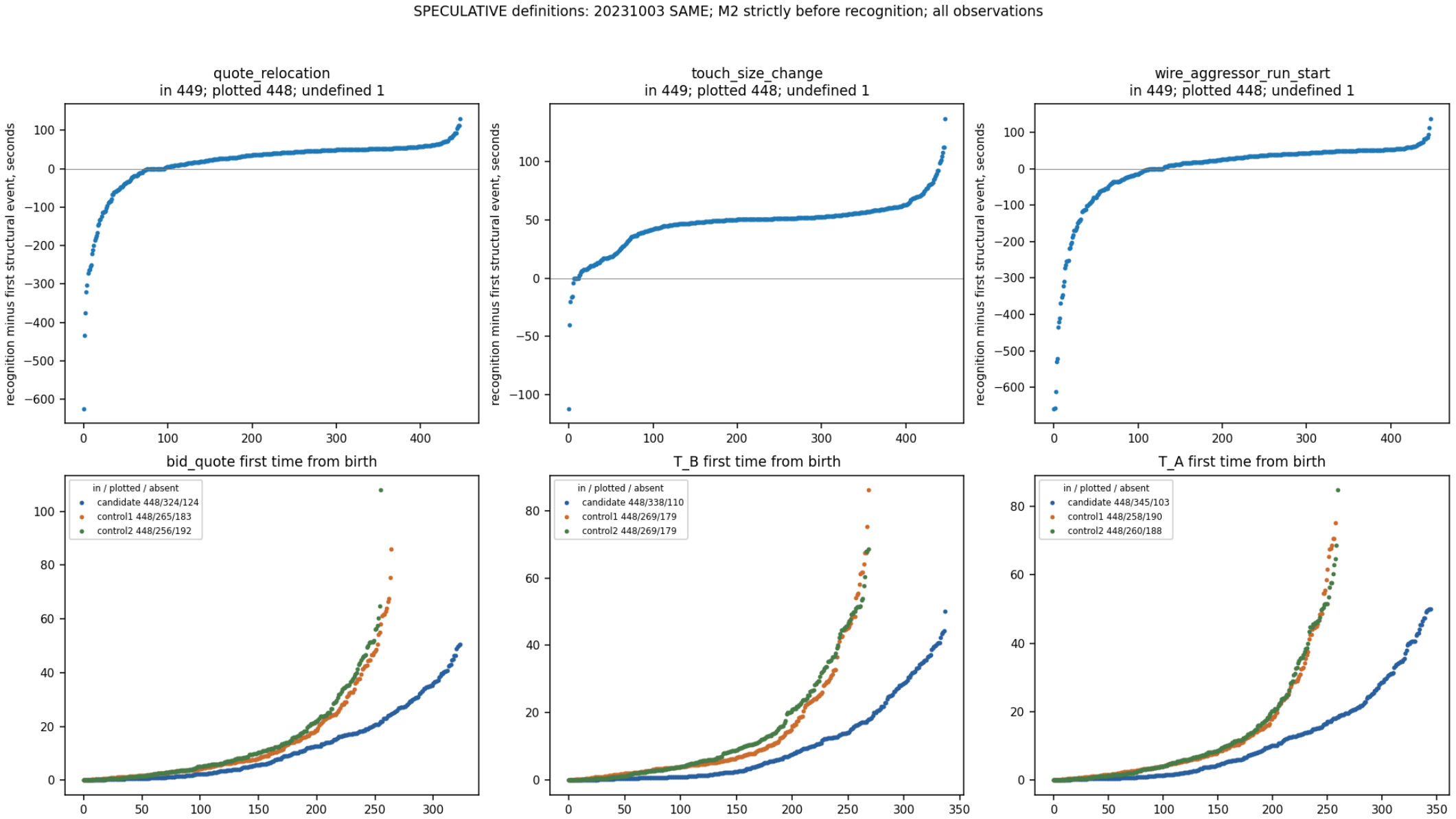
20231003-lag-distributions.png

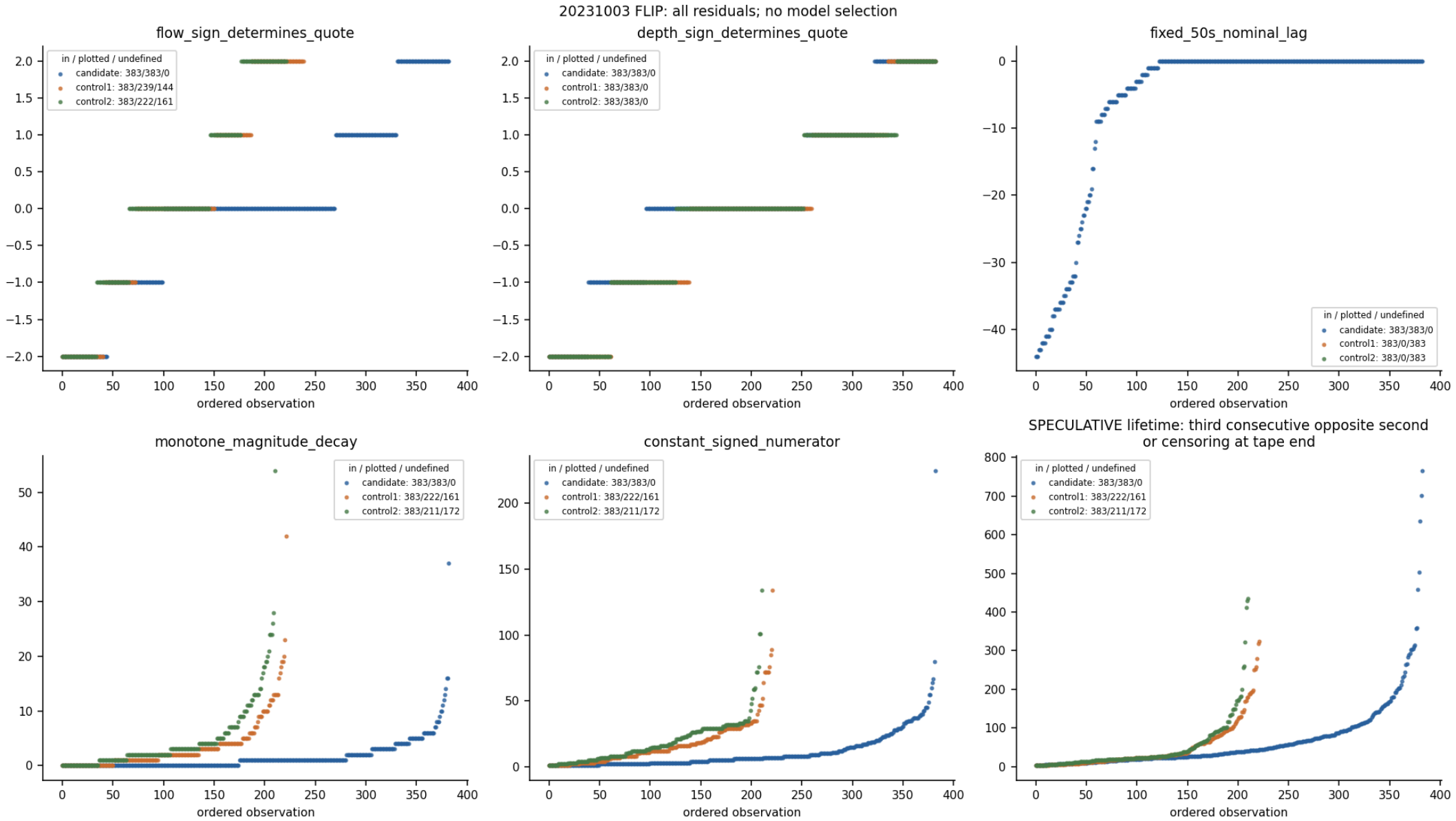
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

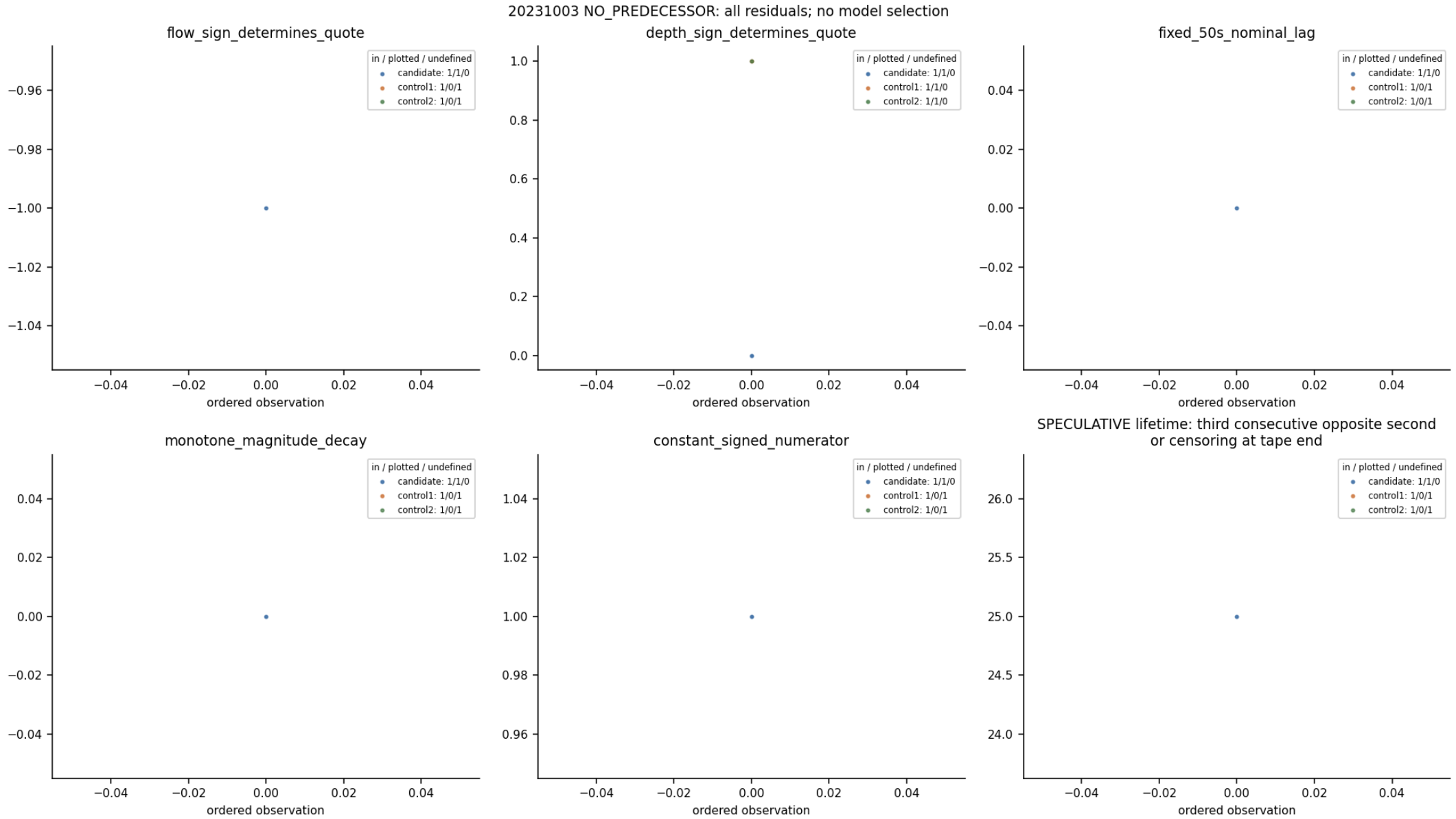


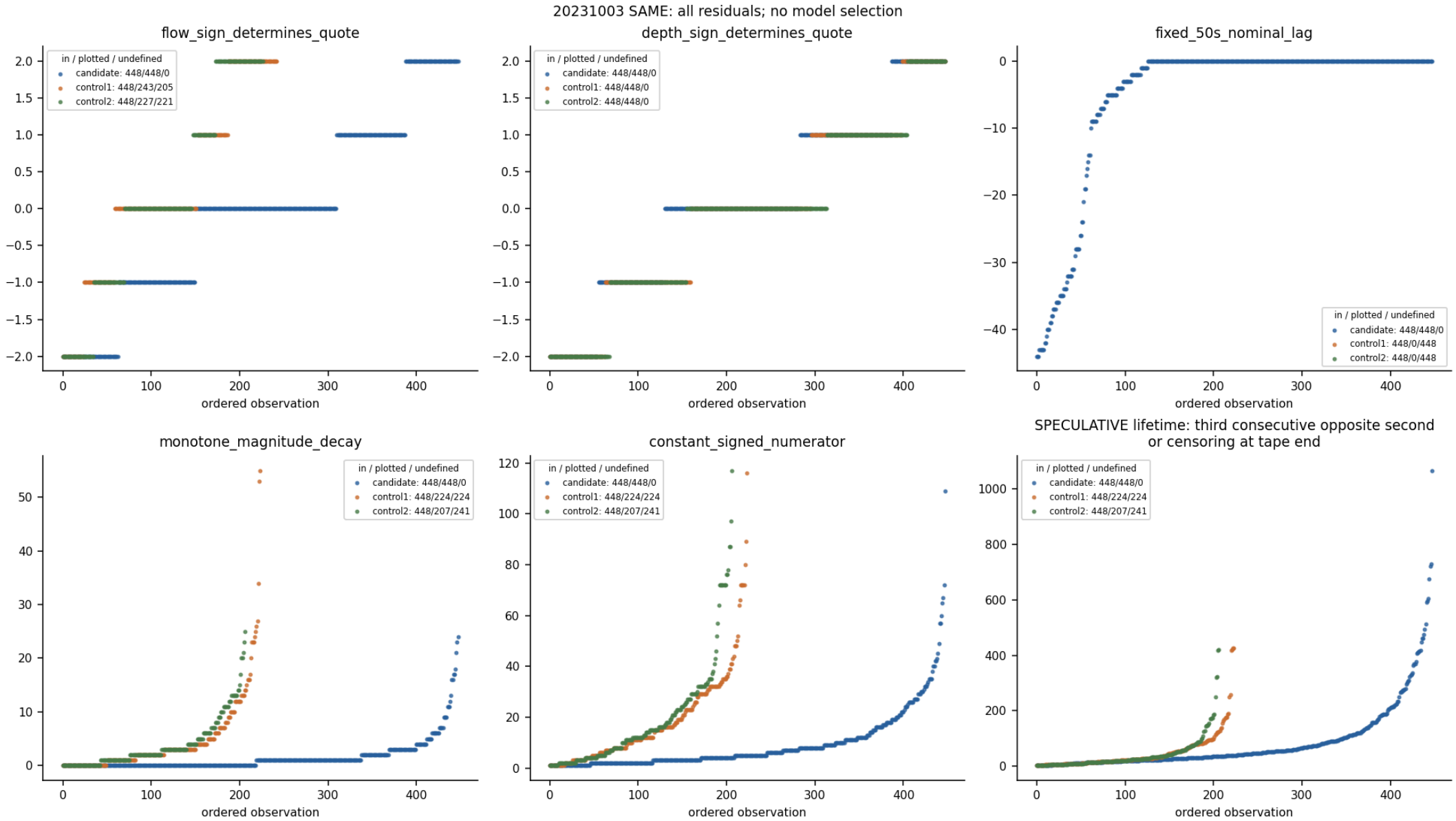

SPECULATIVE definitions: 20231003 FLIP; M2 strictly before recognition; all observations
quote_relocation
in 383; plotted 383; undefined 0
touch_size_change
in 383; plotted 383; undefined 0
wire_aggressor_run_start
in 383; plotted 383; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 383/284/99
control1 383/223/160
control2 383/234/149
candidate 383/291/92
control1 383/235/148
control2 383/235/148
candidate 383/299/84
control1 383/233/150
control2 383/232/151

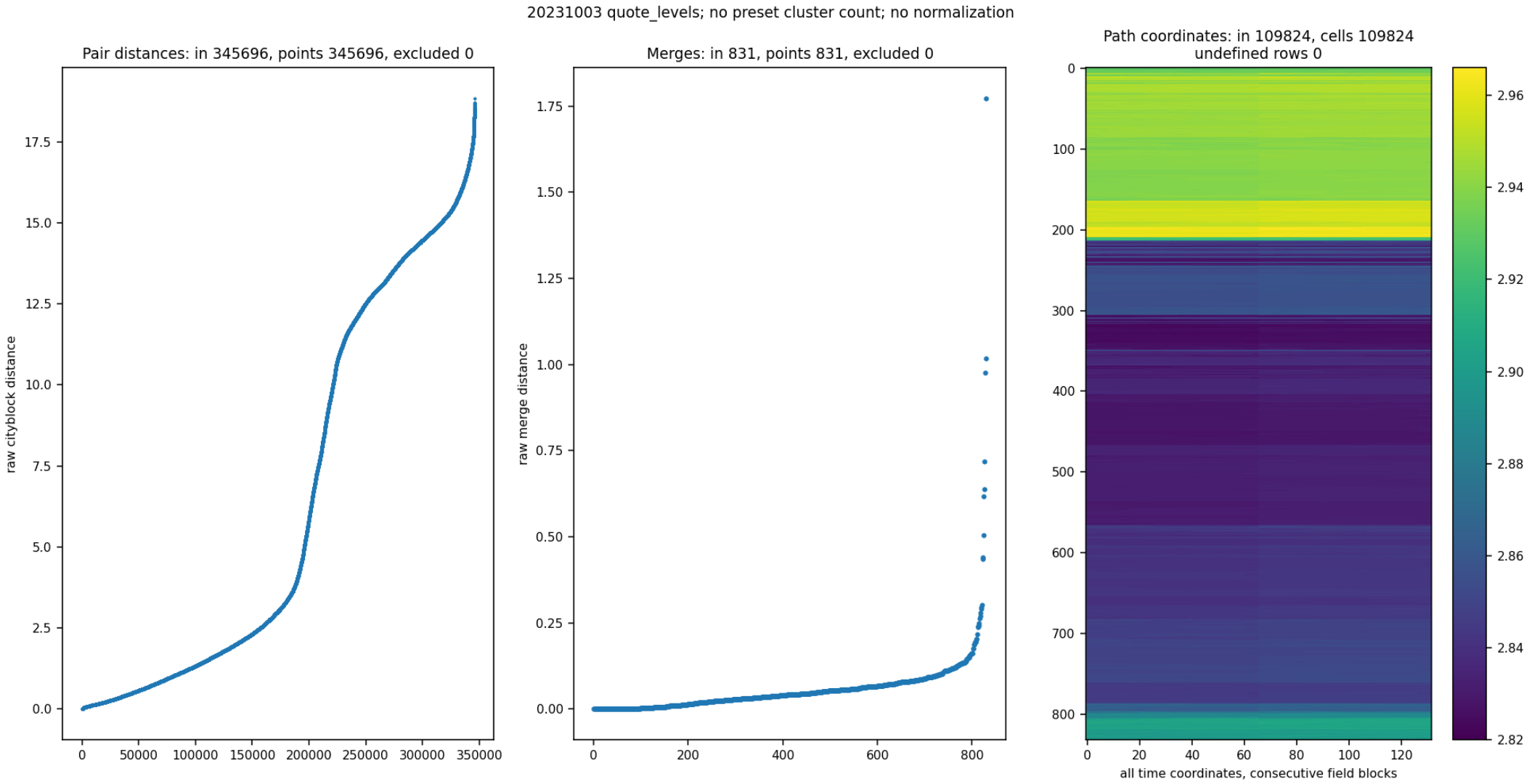

SPECULATIVE definitions: 20231003 NO_PREDECESSOR; M2 strictly before recognition; all observations
quote_relocation
in 1; plotted 1; undefined 0
touch_size_change
in 1; plotted 1; undefined 0
wire_aggressor_run_start
in 1; plotted 1; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 1/1/0
control1 1/0/1
control2 1/0/1

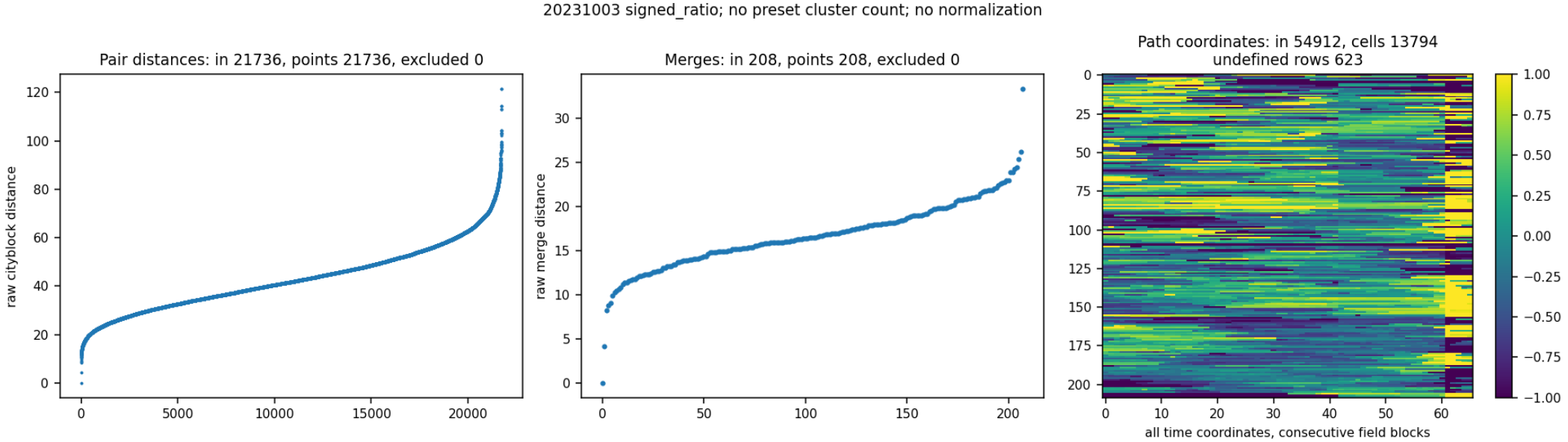

SPECULATIVE definitions: 20231003 SAME; M2 strictly before recognition; all observations
quote_relocation
in 449; plotted 448; undefined 1
touch_size_change
in 449; plotted 448; undefined 1
wire_aggressor_run_start
in 449; plotted 448; undefined 1
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 448/324/124
control1 448/265/183
control2 448/256/192
candidate 448/338/110
control1 448/269/179
control2 448/269/179
candidate 448/345/103
control1 448/258/190
control2 448/260/188

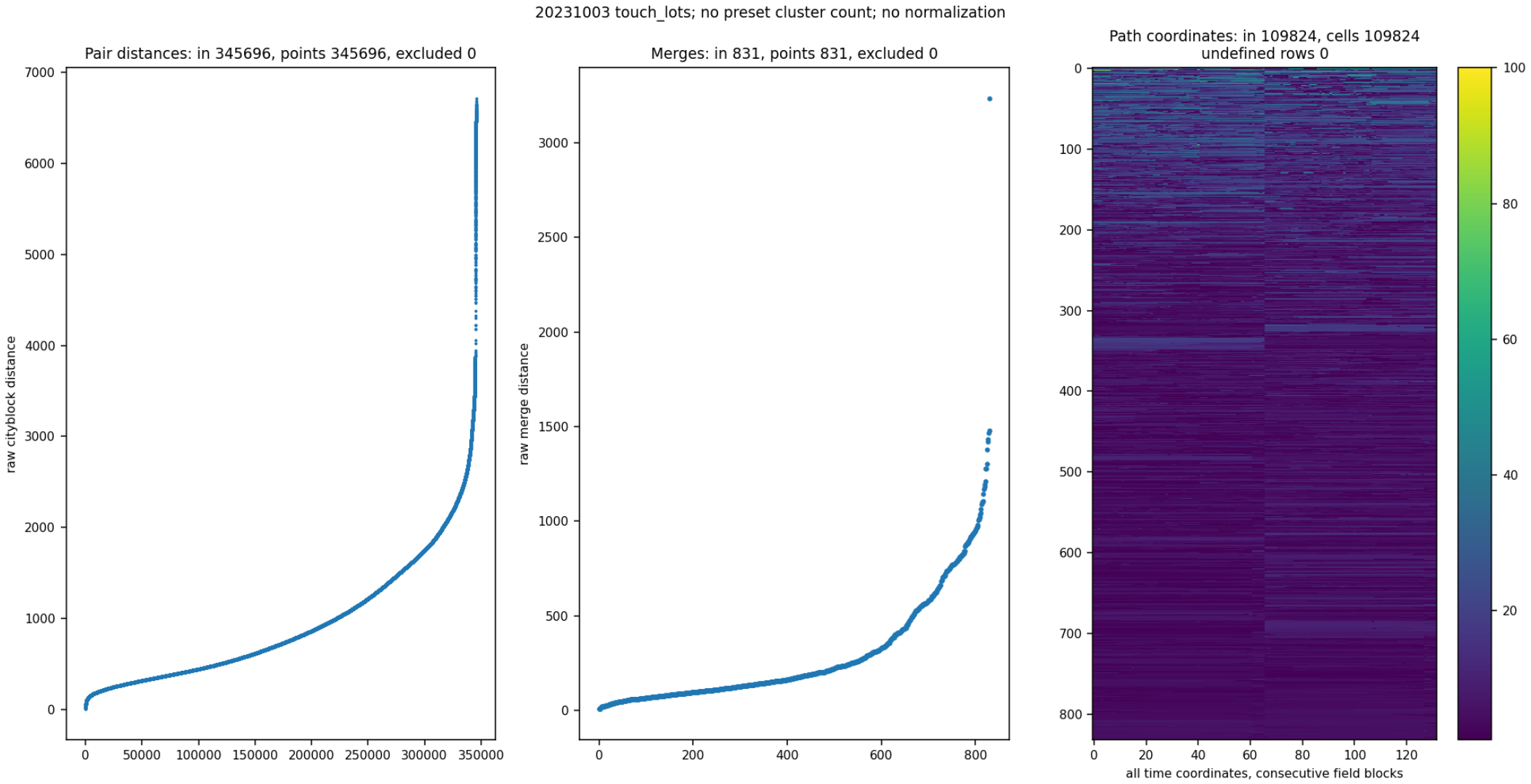

20231003 FLIP: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 383/383/0
control1: 383/239/144
control2: 383/222/161
control1: 383/383/0
control2: 383/383/0
control1: 383/0/383
control2: 383/0/383
control1: 383/222/161
control2: 383/211/172
ordered observation


20231003 NO_PREDECESSOR: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 1/1/0
control1: 1/0/1
control2: 1/0/1
control1: 1/1/0
control2: 1/1/0
ordered observation


20231003 SAME: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 448/448/0
control1: 448/243/205
control2: 448/227/221
control1: 448/448/0
control2: 448/448/0
control1: 448/0/448
control2: 448/0/448
control1: 448/224/224
control2: 448/207/241
ordered observation


20231003 quote_levels; no preset cluster count; no normalization
Pair distances: in 345696, points 345696, excluded 0
raw cityblock distance
Merges: in 831, points 831, excluded 0
raw merge distance
Path coordinates: in 109824, cells 109824
undefined rows 0
all time coordinates, consecutive field blocks


20231003 signed_ratio; no preset cluster count; no normalization
Pair distances: in 21736, points 21736, excluded 0
raw cityblock distance
Merges: in 208, points 208, excluded 0
raw merge distance
Path coordinates: in 54912, cells 13794
undefined rows 623
all time coordinates, consecutive field blocks


20231003 touch_lots; no preset cluster count; no normalization
Pair distances: in 345696, points 345696, excluded 0
raw cityblock distance
Merges: in 831, points 831, excluded 0
raw merge distance
Path coordinates: in 109824, cells 109824
undefined rows 0
all time coordinates, consecutive field blocks

20231003-structure-wire_trade_lots.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

